

BFS Traversal

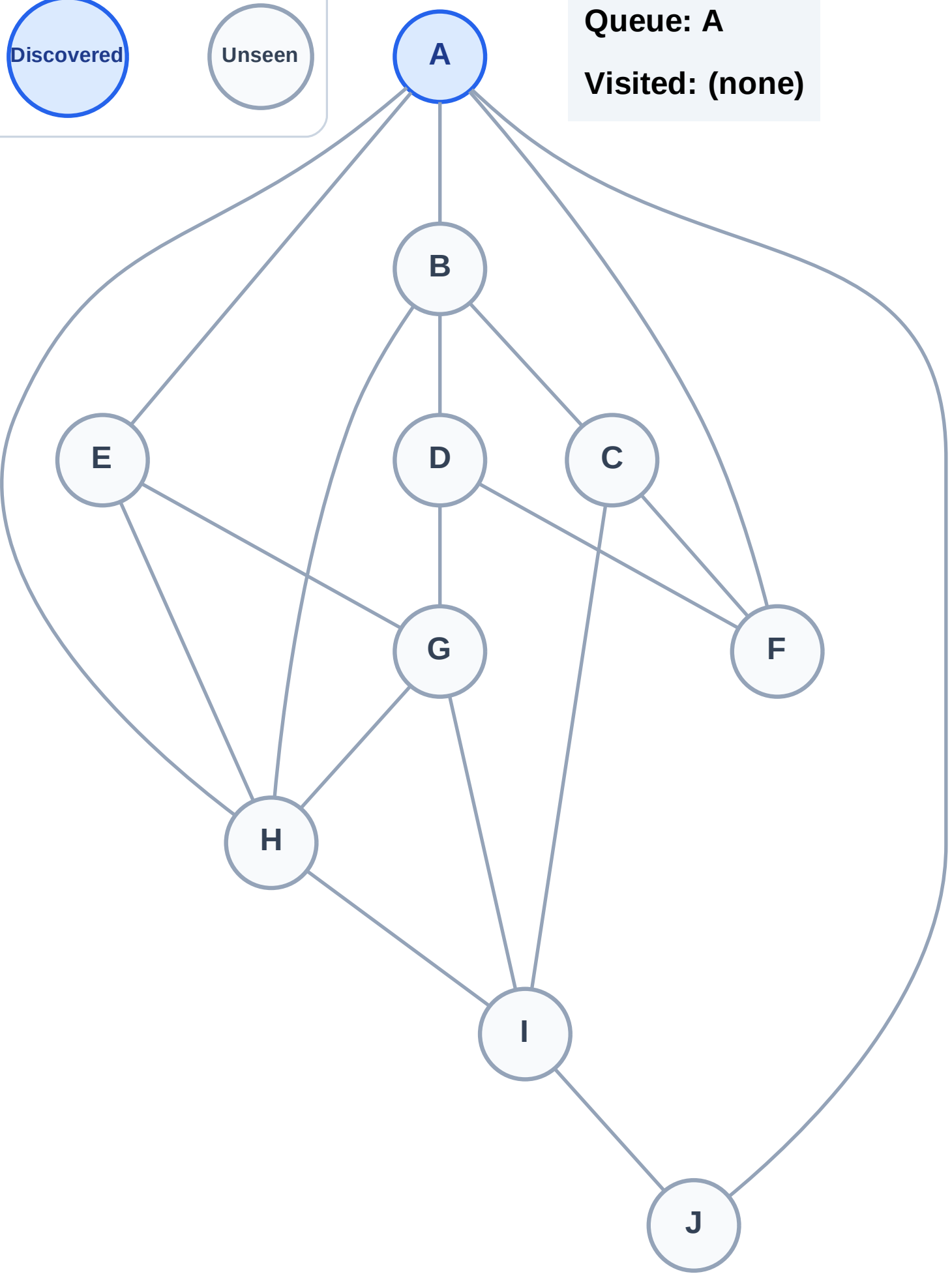
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

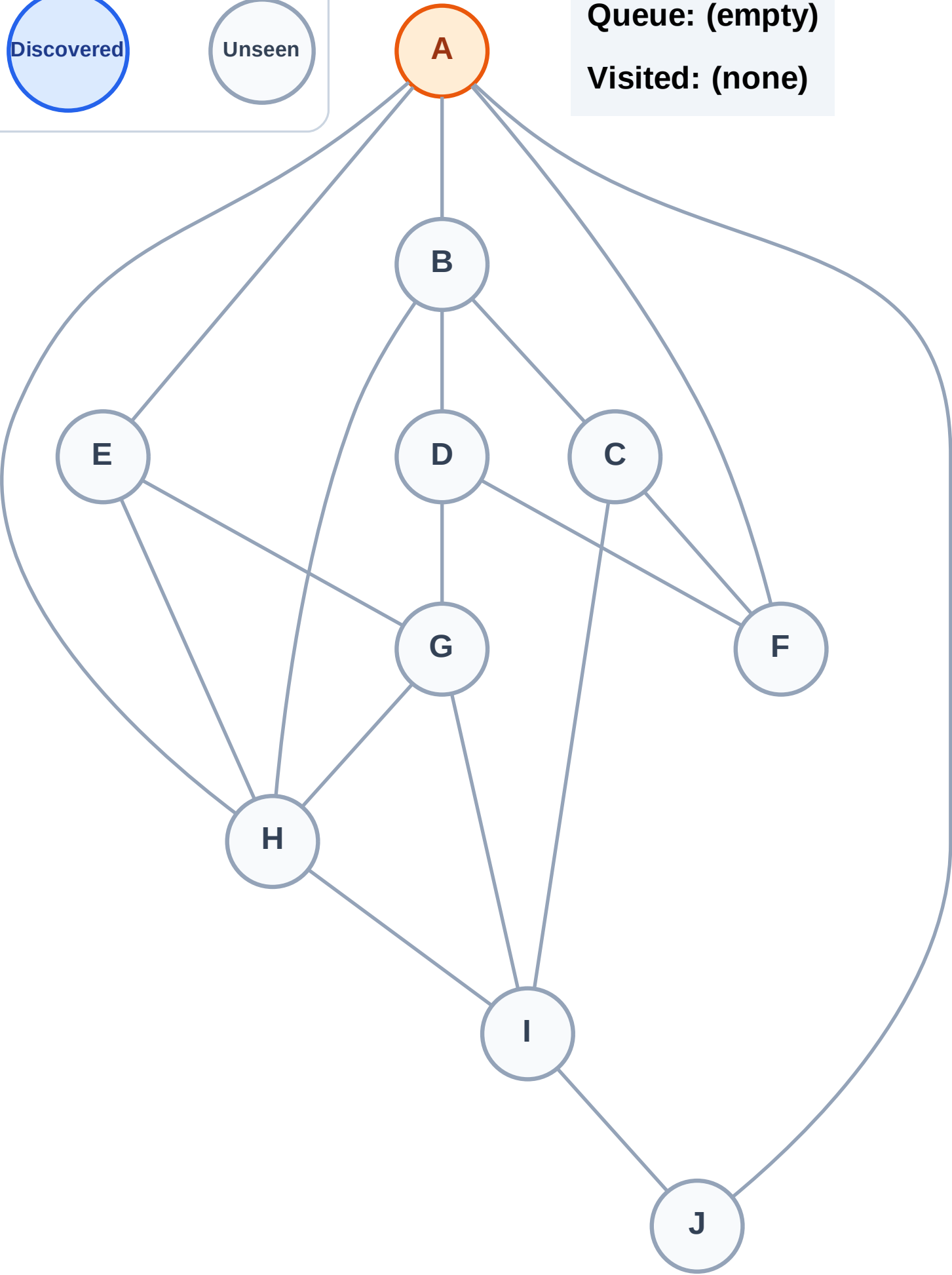
Complete

Processing

Discovered

Unseen

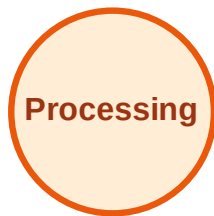
Queue: (empty)
Visited: (none)



BFS Traversal

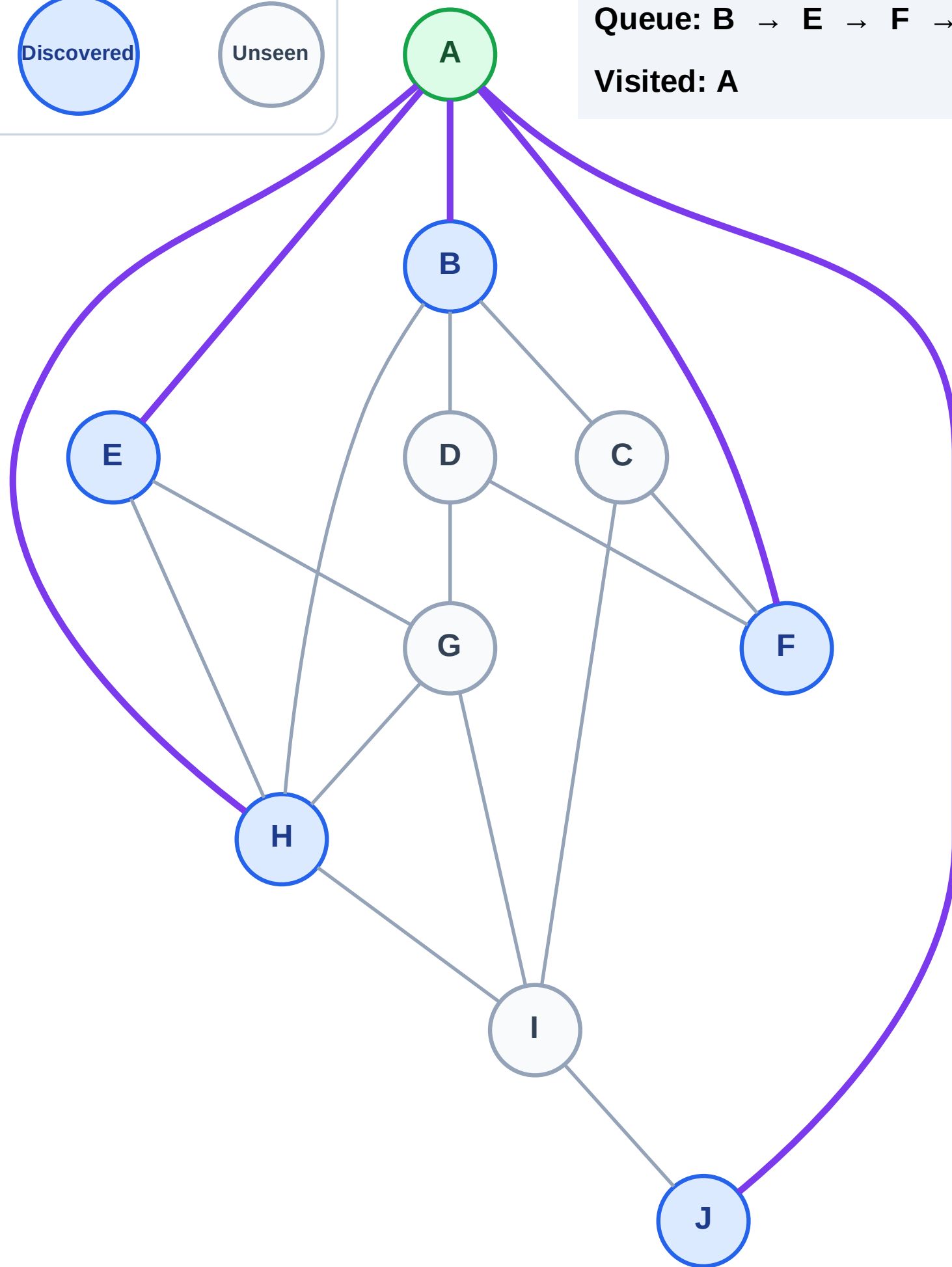
Step 3: Discover B, E, F, H, J from A

Legend



Queue: B → E → F → H → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

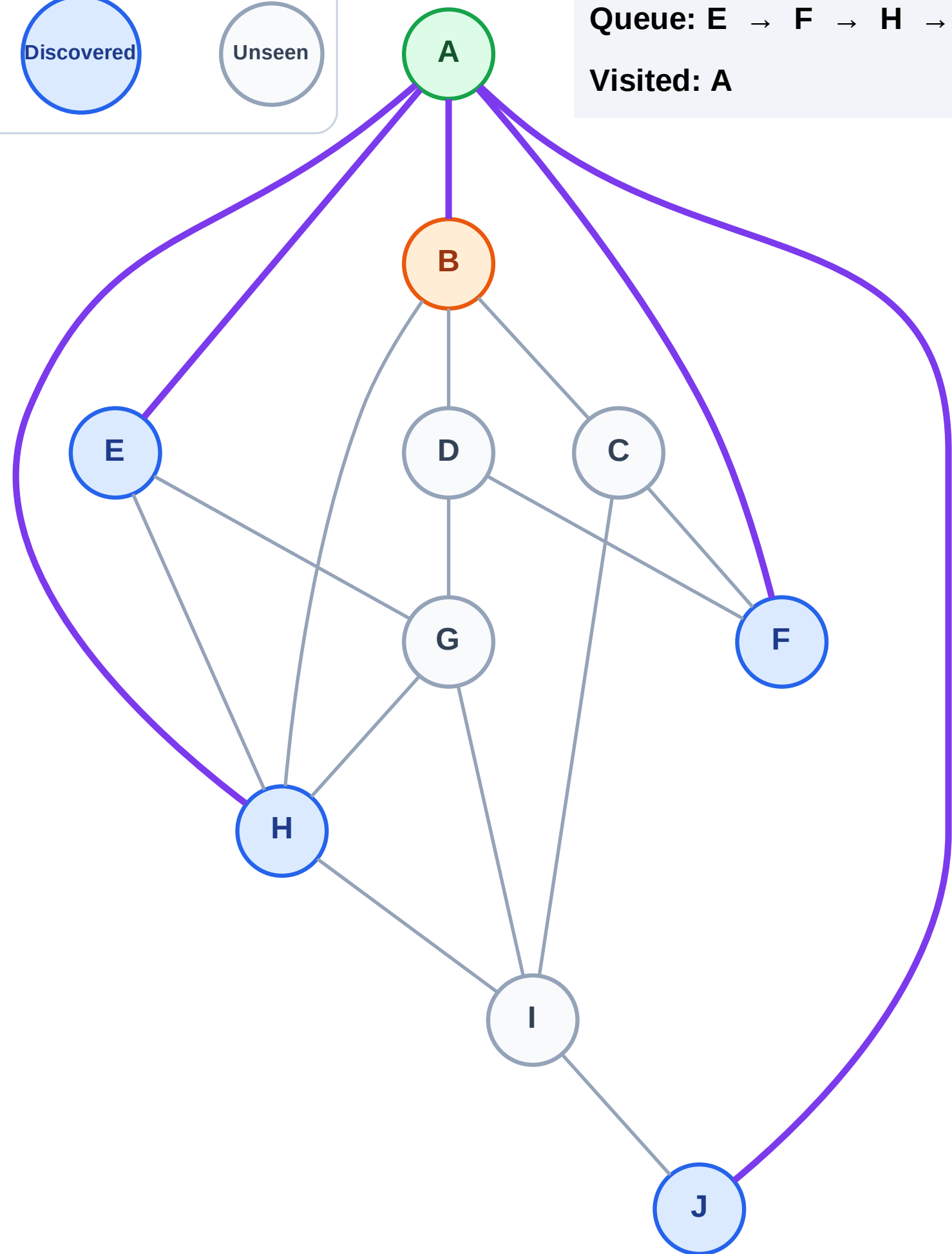
Processing

Discovered

Unseen

Queue: E → F → H → J

Visited: A



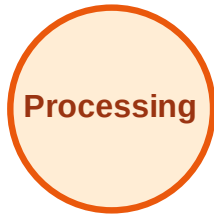
BFS Traversal

Step 5: Discover C, D from B

Legend



Complete



Processing



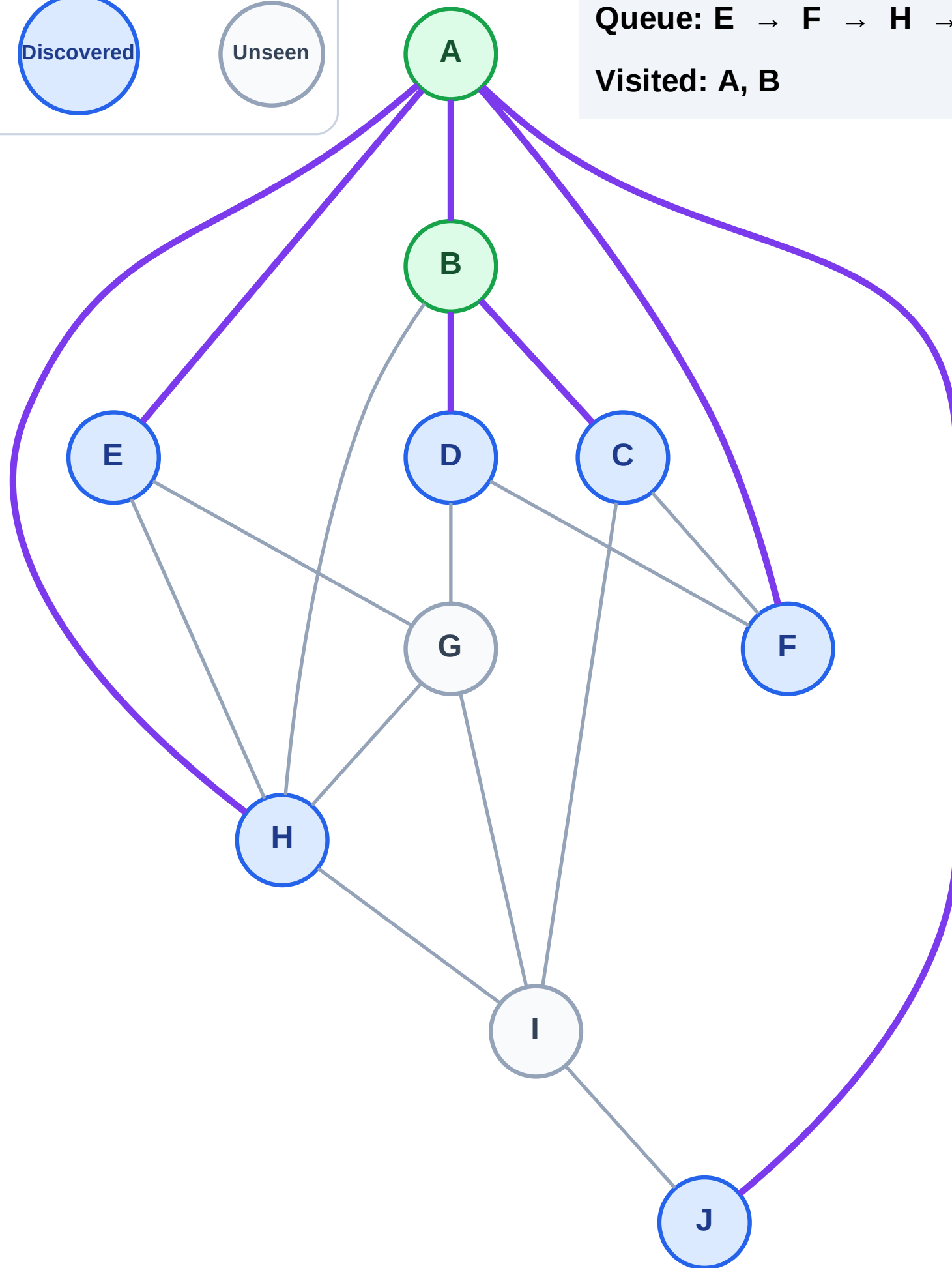
Discovered



Unseen

Queue: E → F → H → J → C → D

Visited: A, B



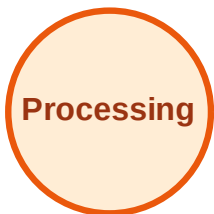
BFS Traversal

Step 6: Process node E

Legend



Complete



Processing



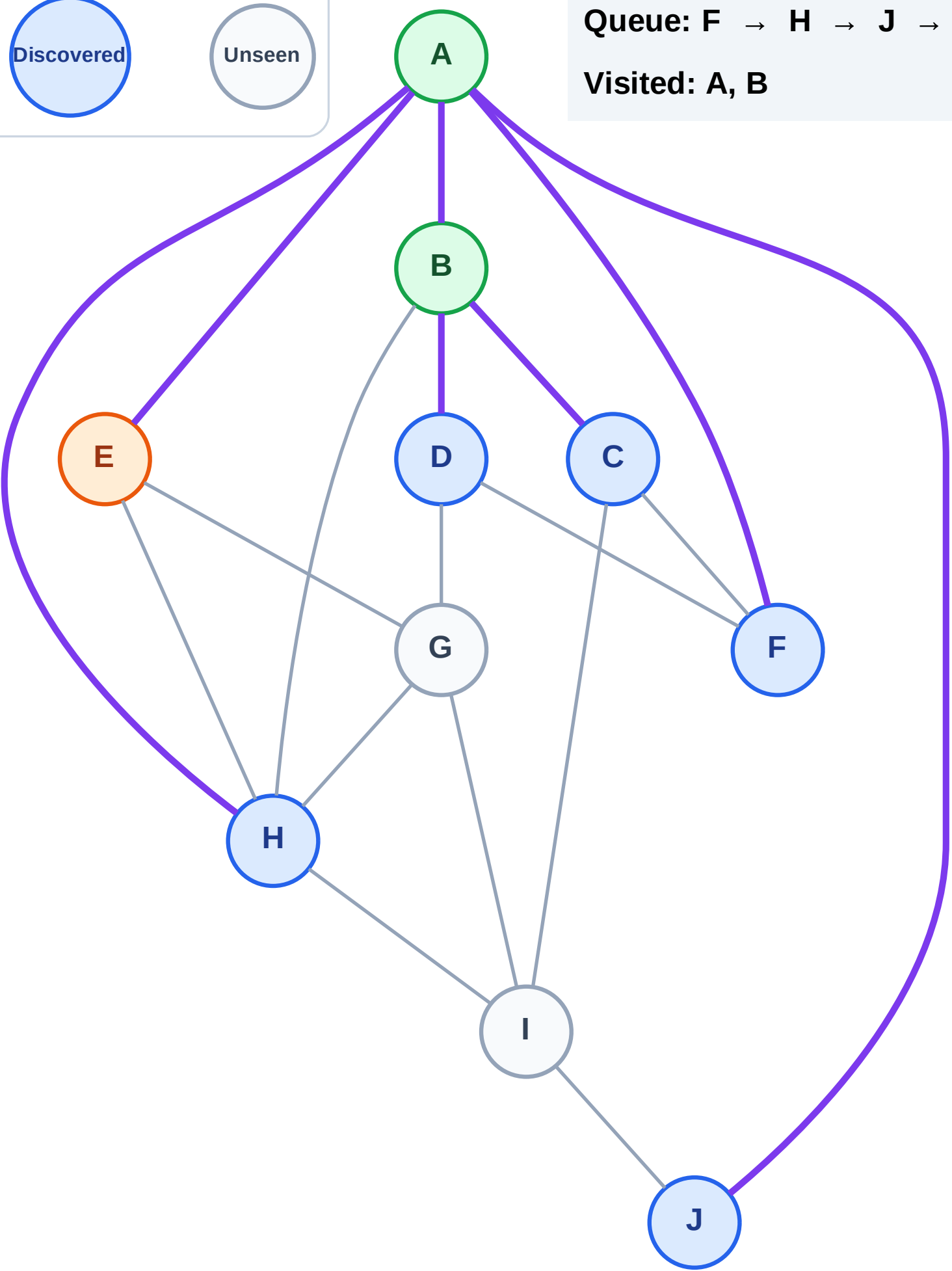
Discovered



Unseen

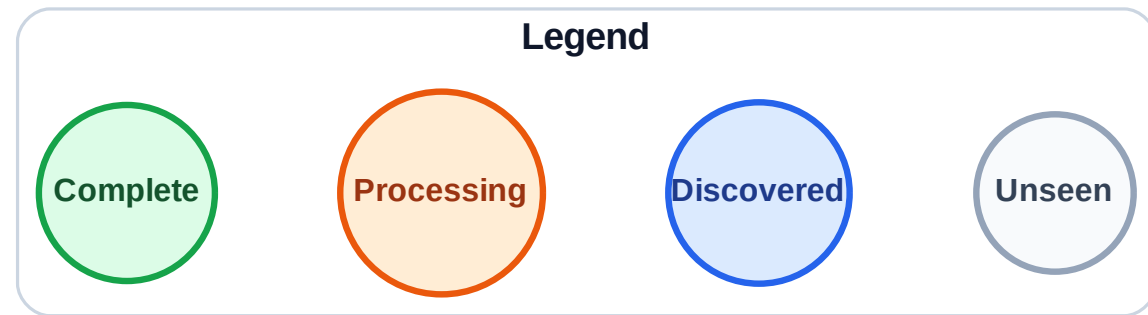
Queue: F → H → J → C → D

Visited: A, B



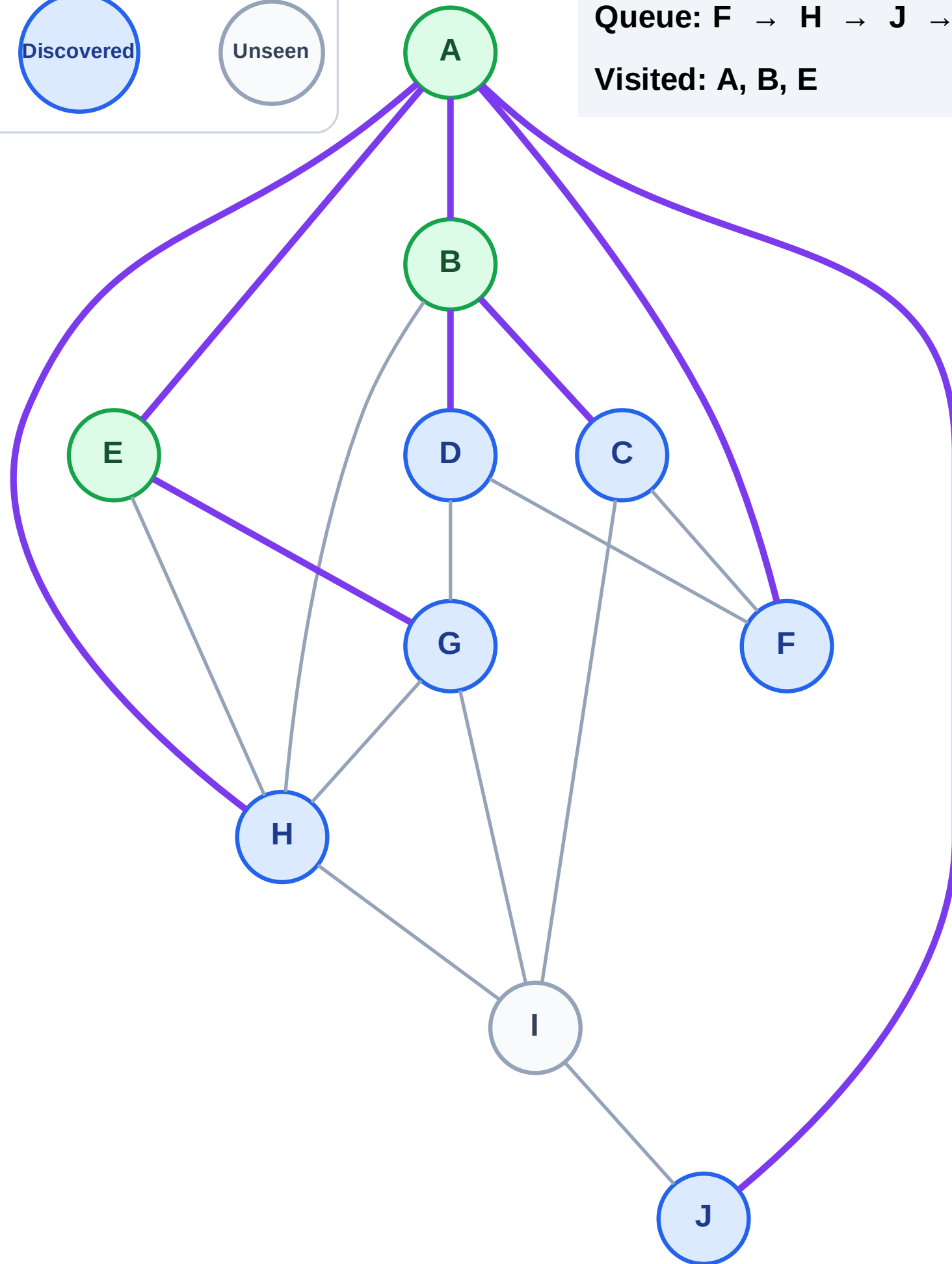
BFS Traversal

Step 7: Discover G from E



Queue: F → H → J → C → D → G

Visited: A, B, E



BFS Traversal

Step 8: Process node F

Legend

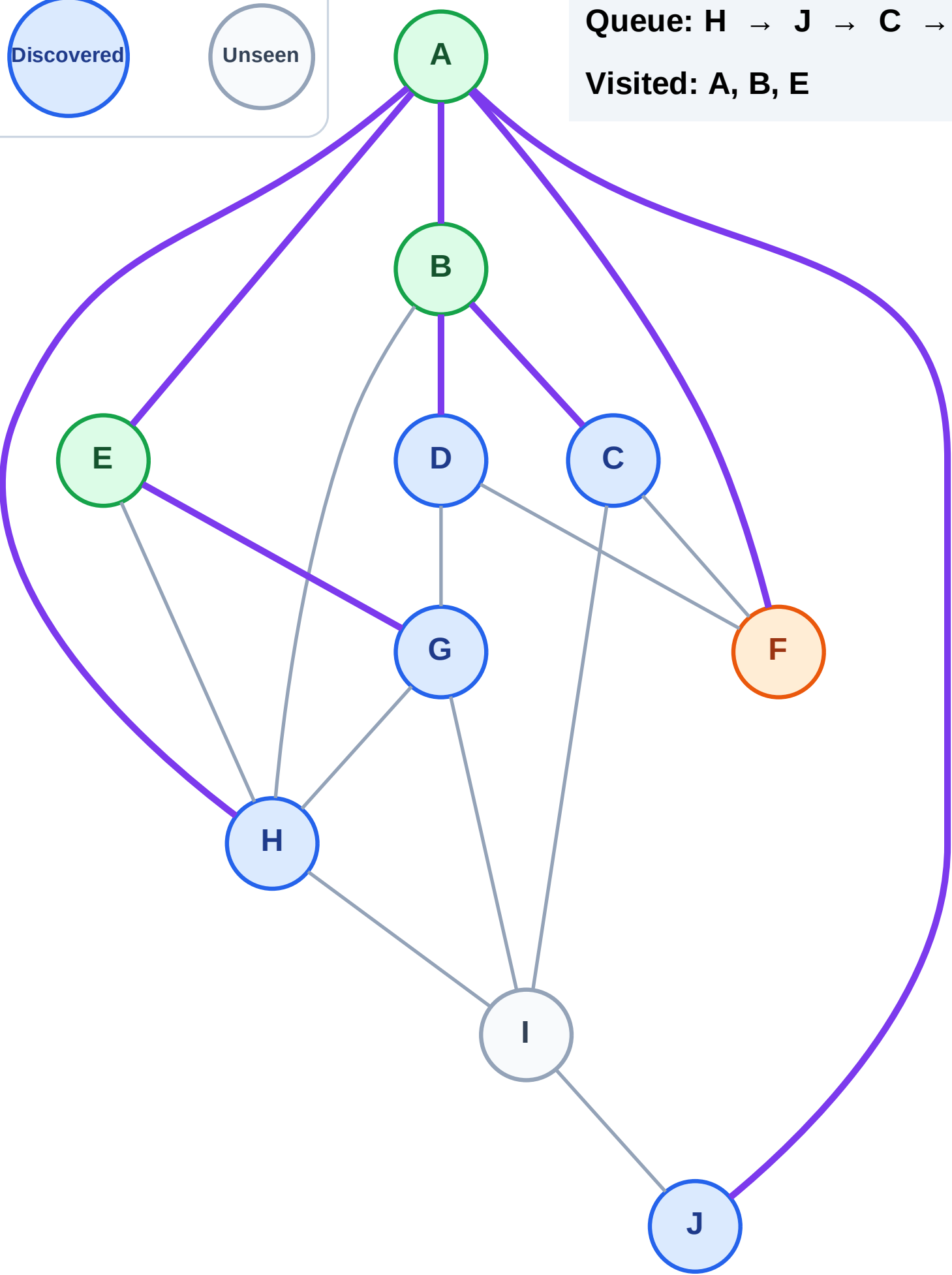


Processing



Queue: H → J → C → D → G

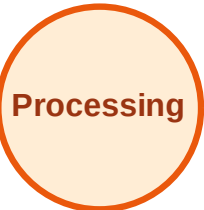
Visited: A, B, E



BFS Traversal

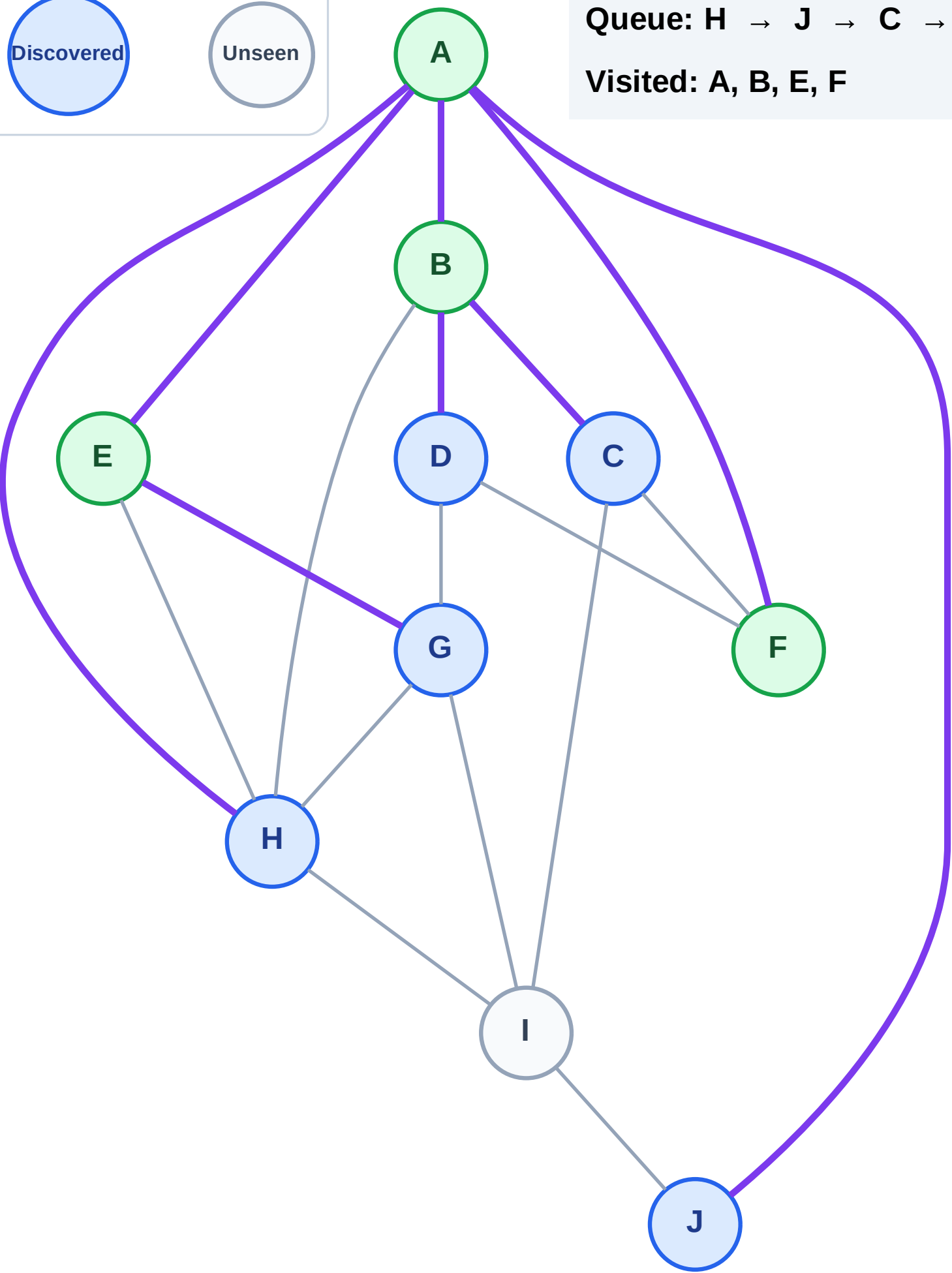
Step 9: No new neighbors from F

Legend



Queue: H → J → C → D → G

Visited: A, B, E, F



BFS Traversal

Step 10: Process node H

Legend

Complete

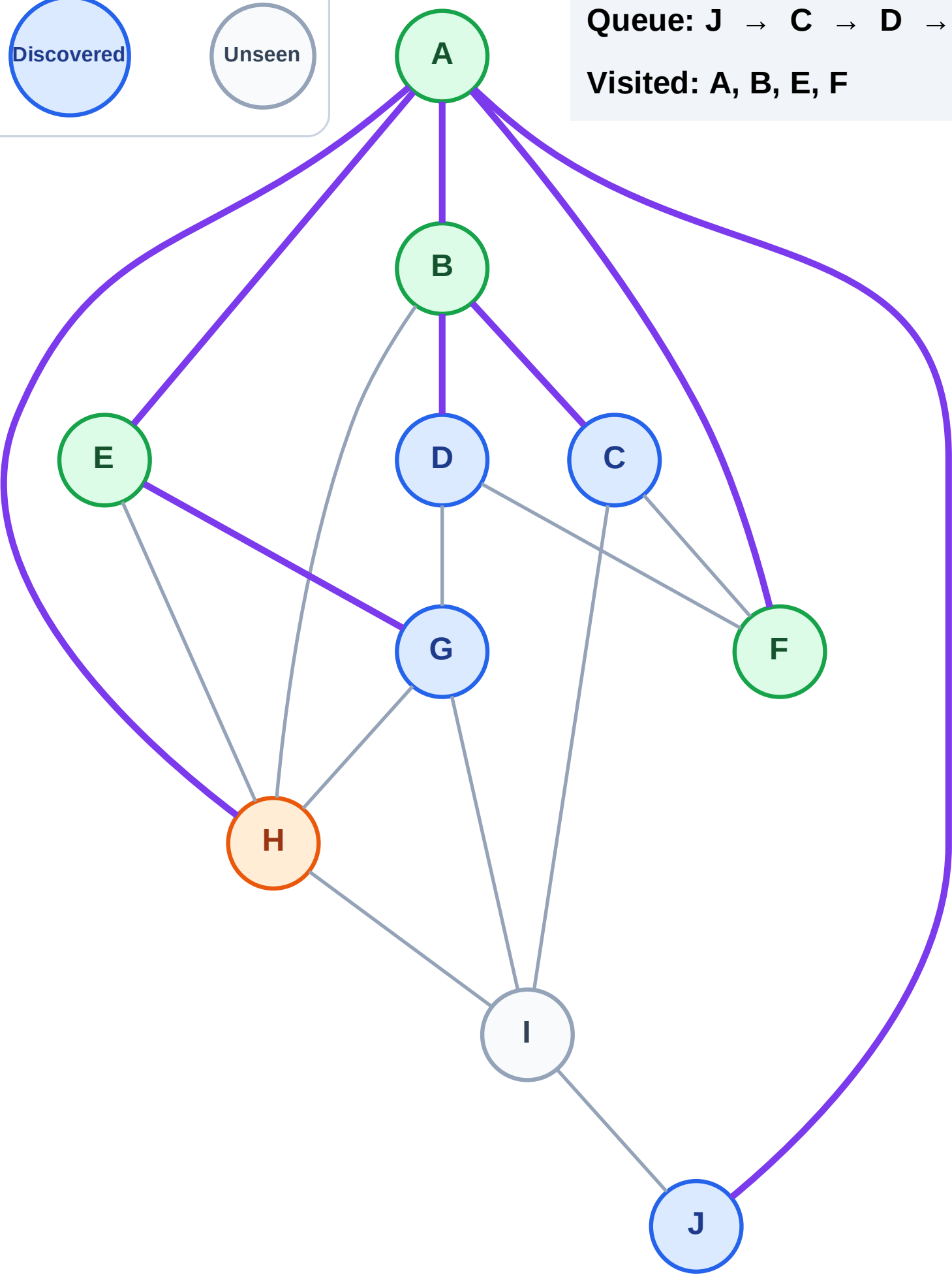
Processing

Discovered

Unseen

Queue: J → C → D → G

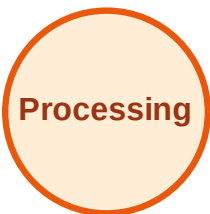
Visited: A, B, E, F



BFS Traversal

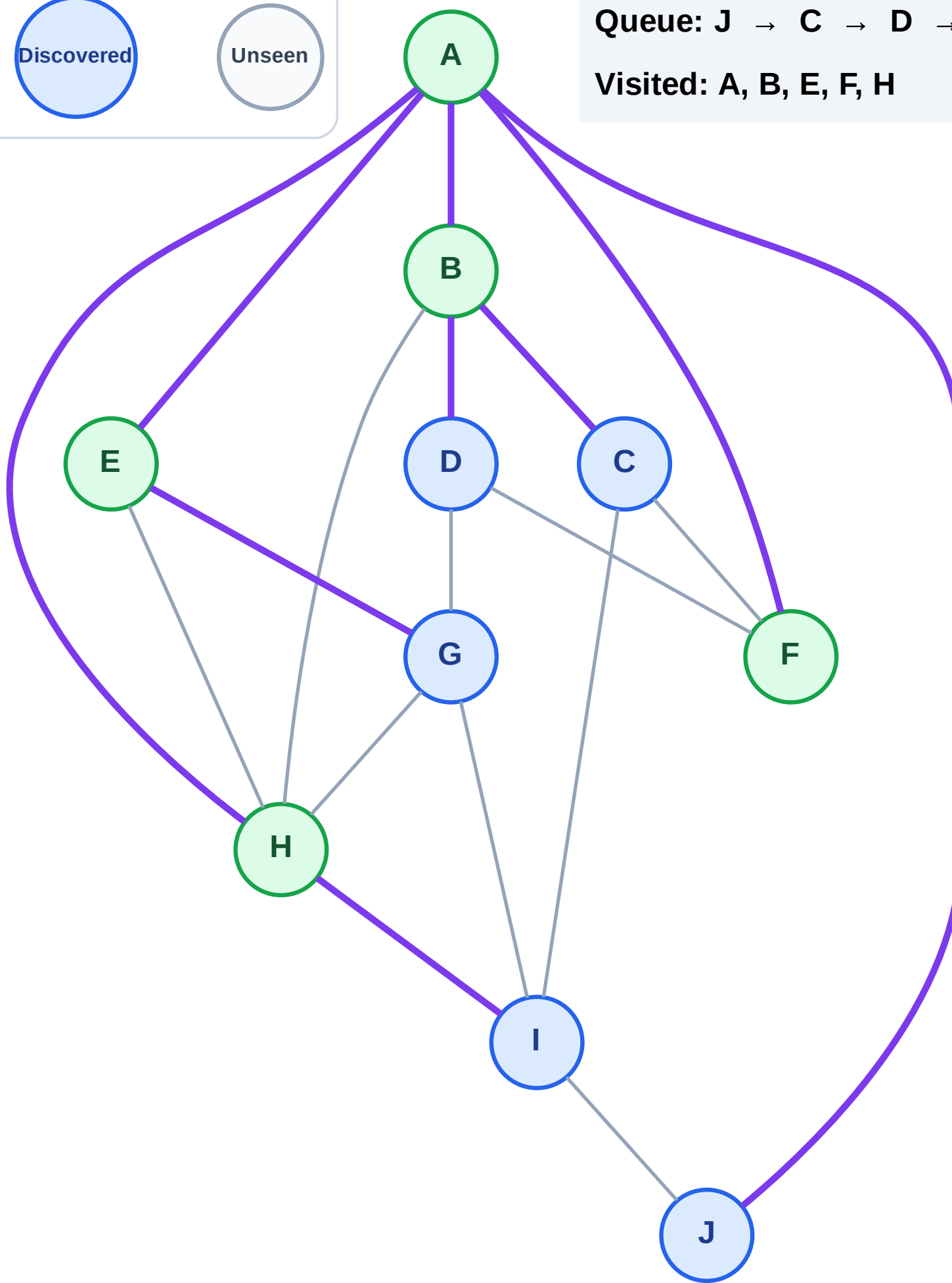
Step 11: Discover I from H

Legend



Queue: J → C → D → G → I

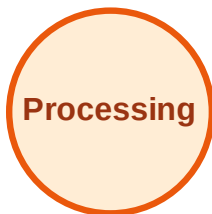
Visited: A, B, E, F, H



BFS Traversal

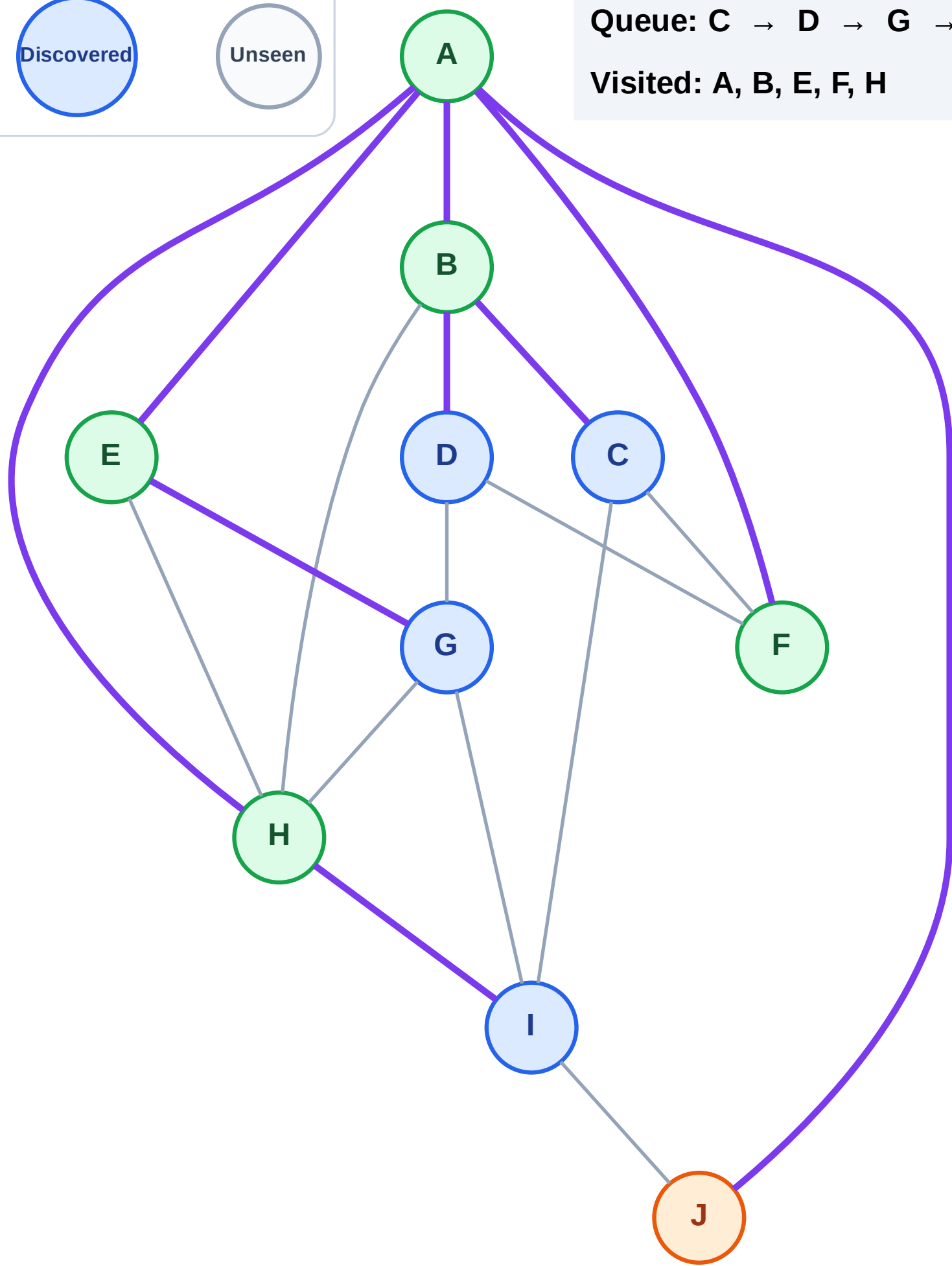
Step 12: Process node J

Legend



Queue: C → D → G → I

Visited: A, B, E, F, H



BFS Traversal

Step 13: No new neighbors from J

Legend

Complete

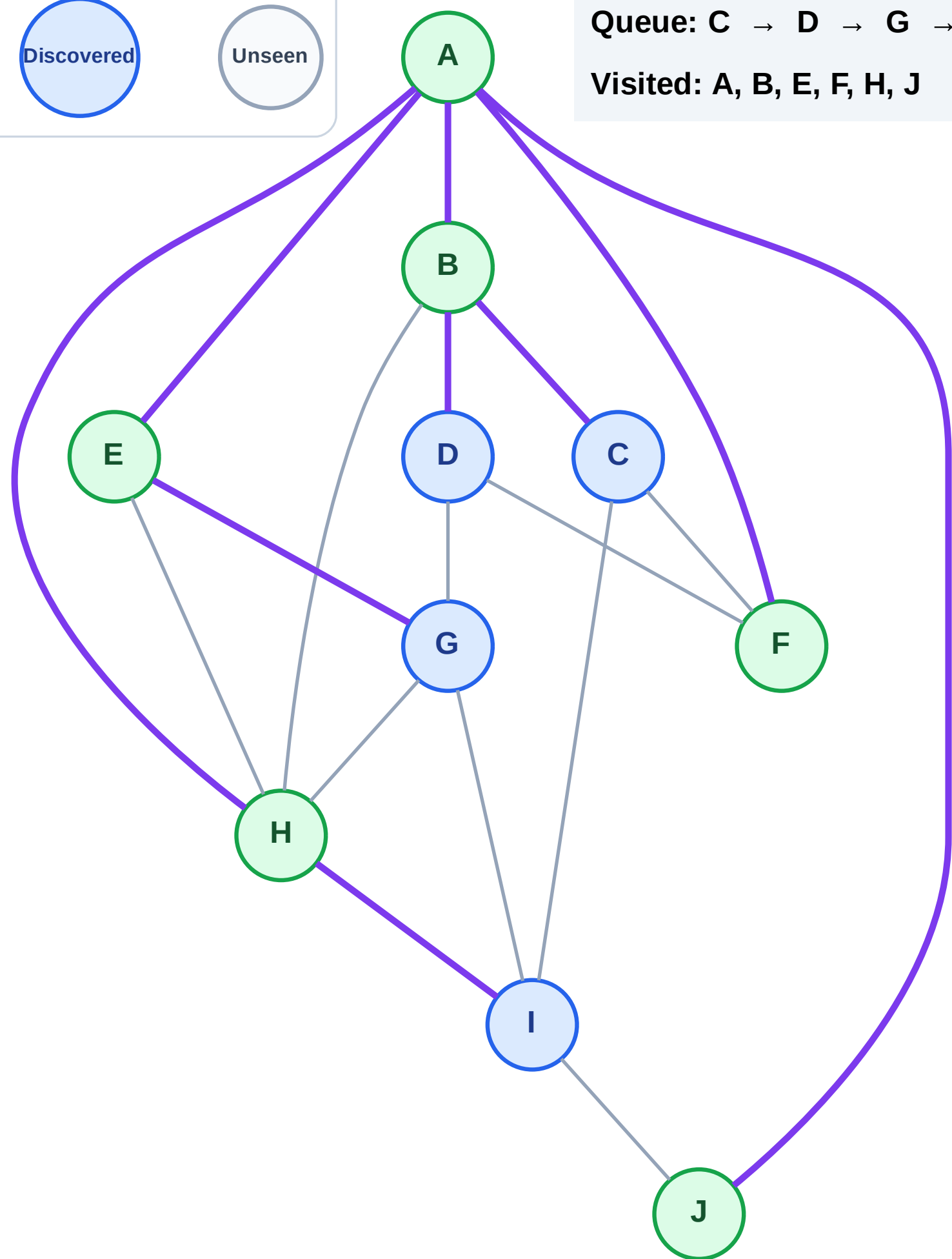
Processing

Discovered

Unseen

Queue: C → D → G → I

Visited: A, B, E, F, H, J



BFS Traversal

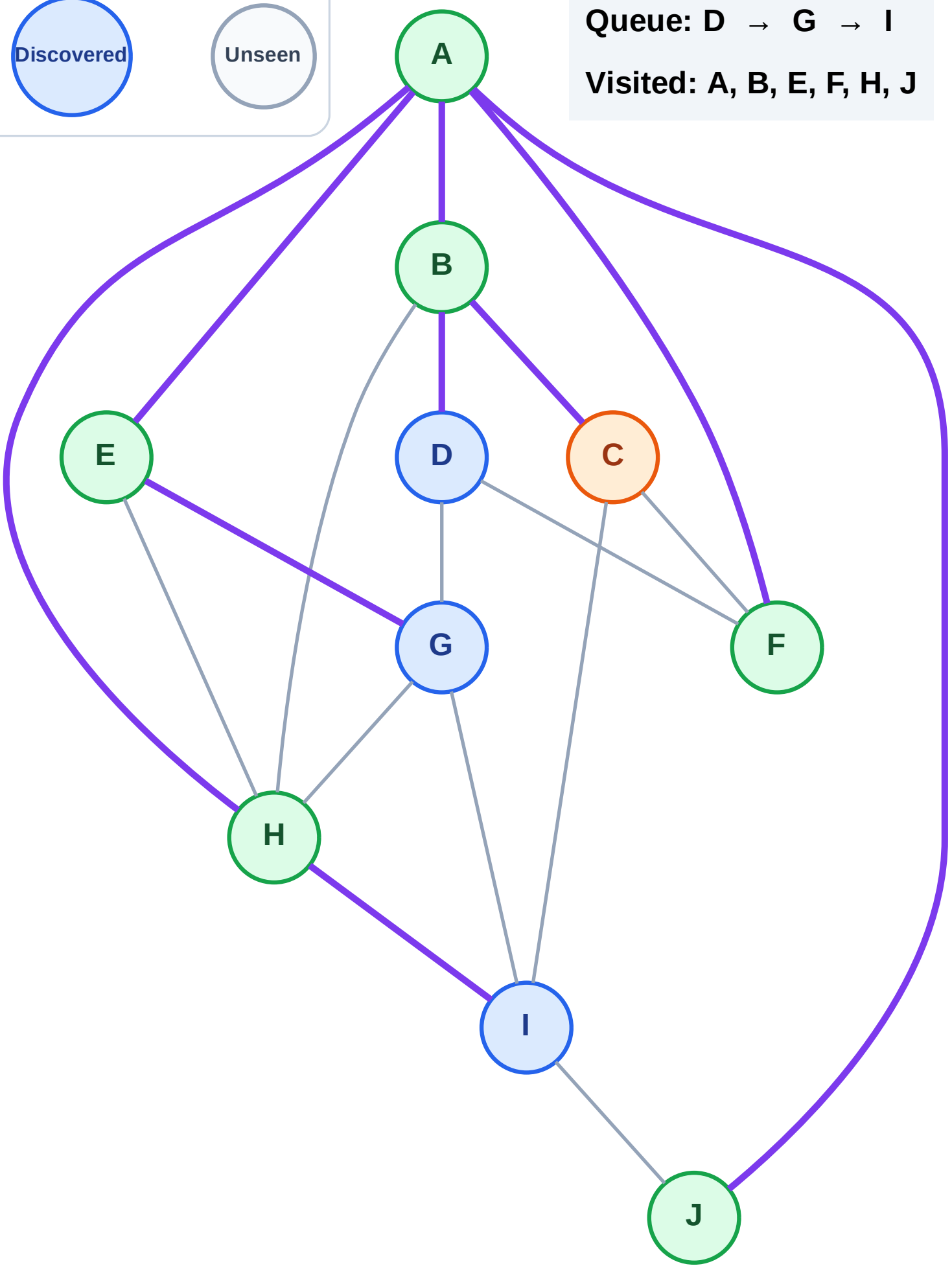
Step 14: Process node C

Legend



Queue: D → G → I

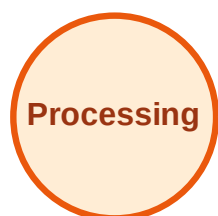
Visited: A, B, E, F, H, J



BFS Traversal

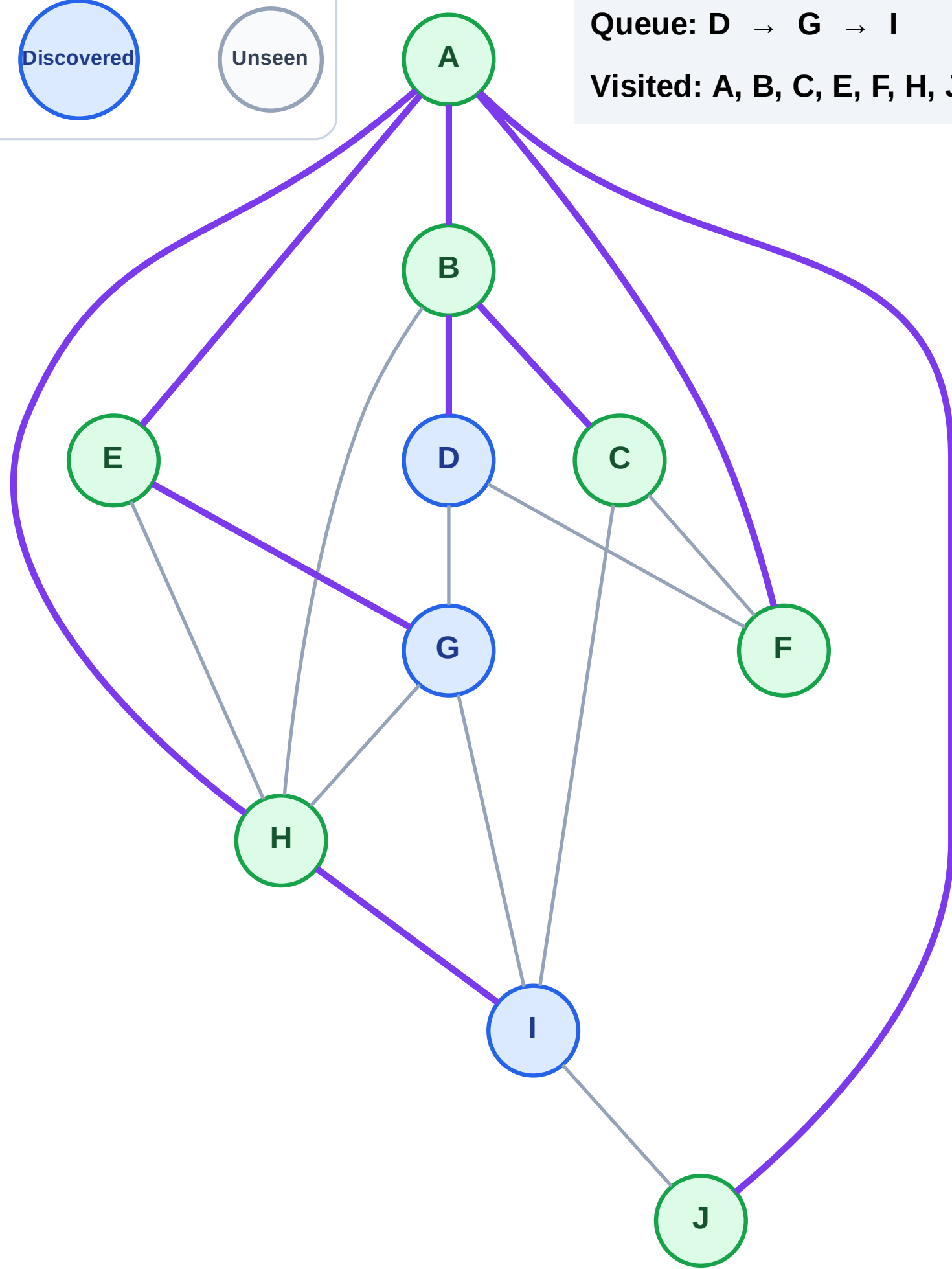
Step 15: No new neighbors from C

Legend



Queue: D → G → I

Visited: A, B, C, E, F, H, J



BFS Traversal

Step 16: Process node D

Legend

Complete

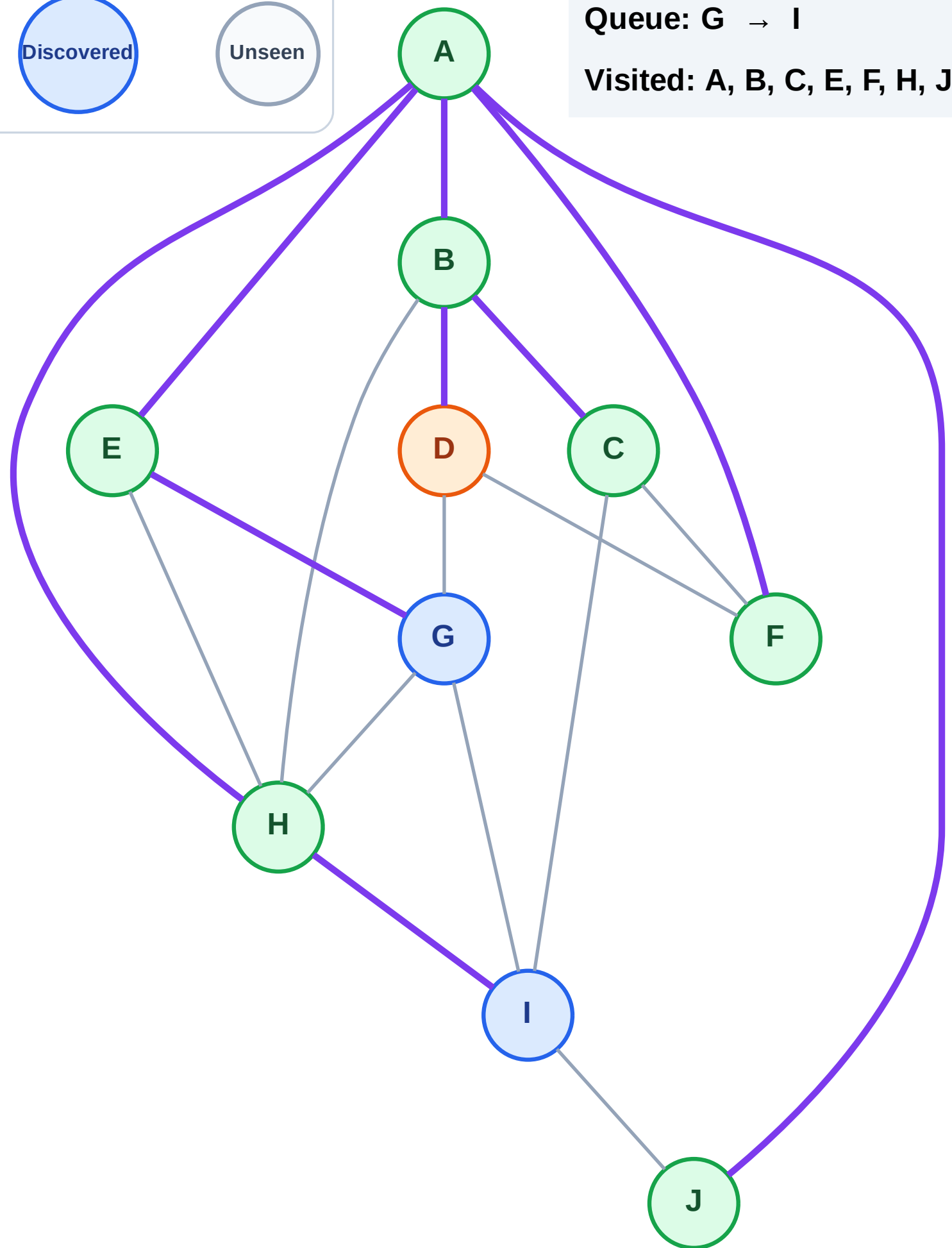
Processing

Discovered

Unseen

Queue: G → I

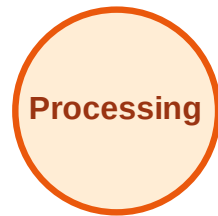
Visited: A, B, C, E, F, H, J



BFS Traversal

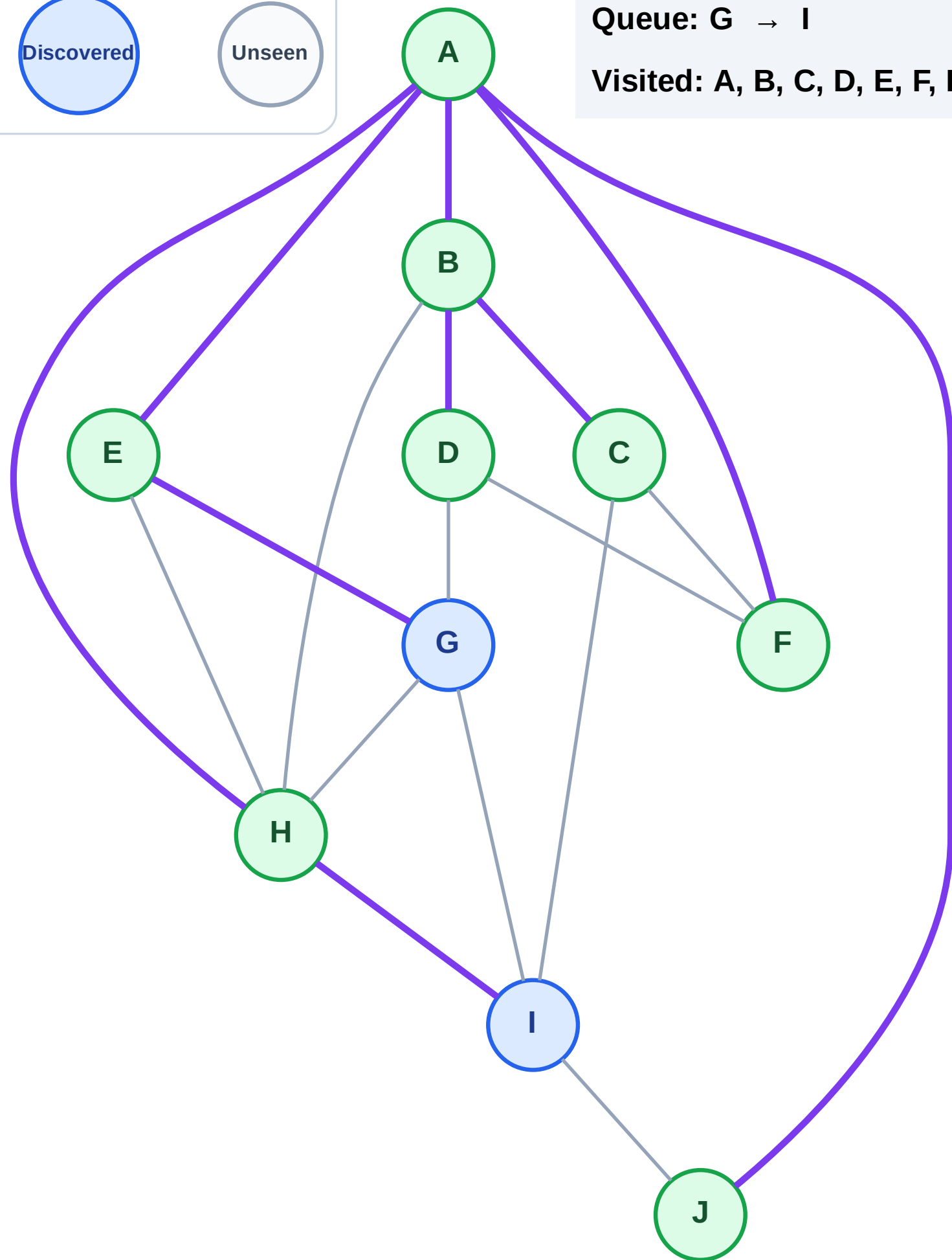
Step 17: No new neighbors from D

Legend



Queue: G → I

Visited: A, B, C, D, E, F, H, J



BFS Traversal

Step 18: Process node G

Legend

Complete

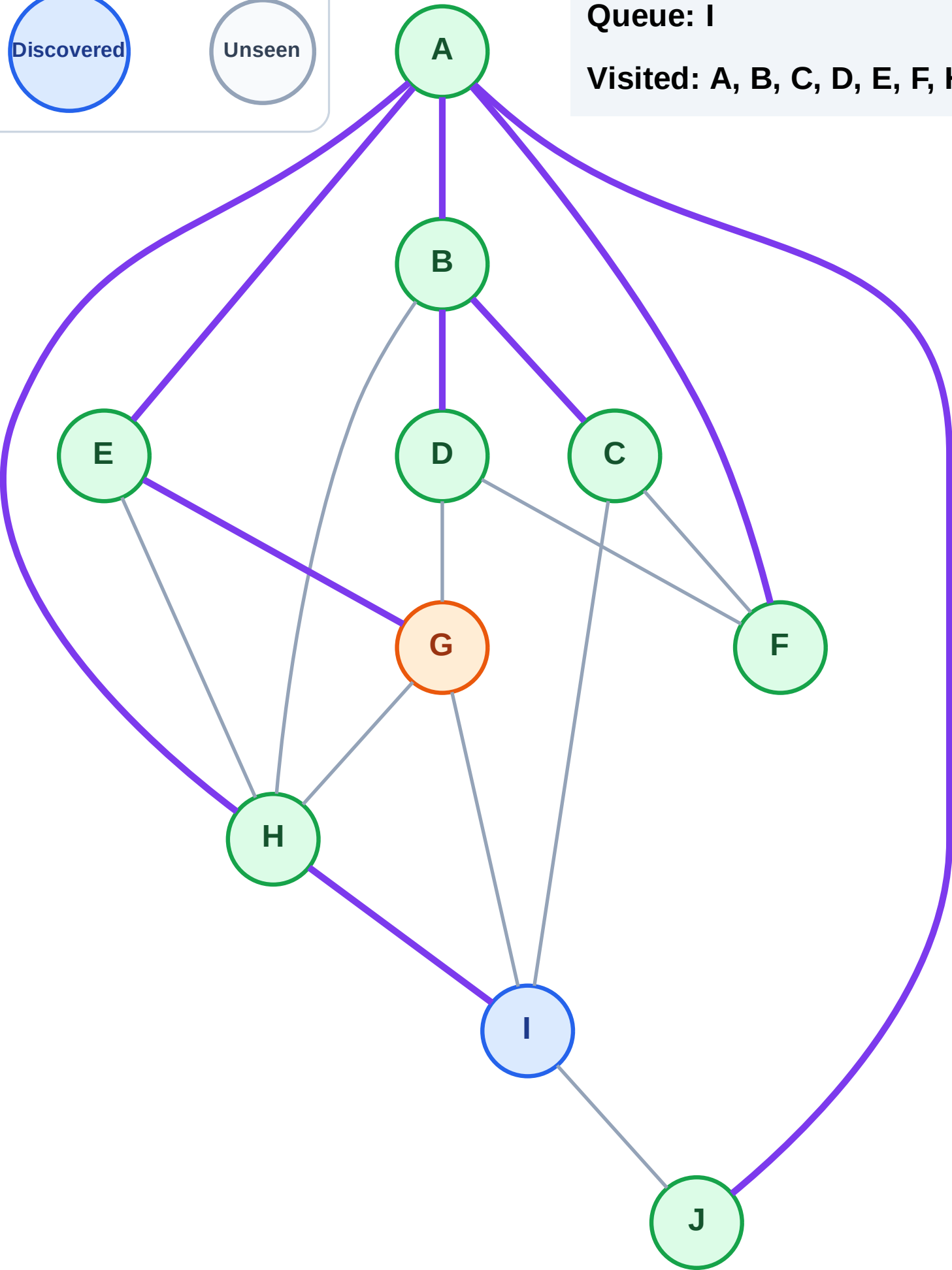
Processing

Discovered

Unseen

Queue: I

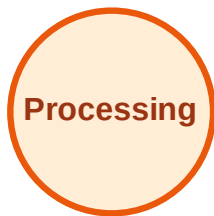
Visited: A, B, C, D, E, F, H, J



BFS Traversal

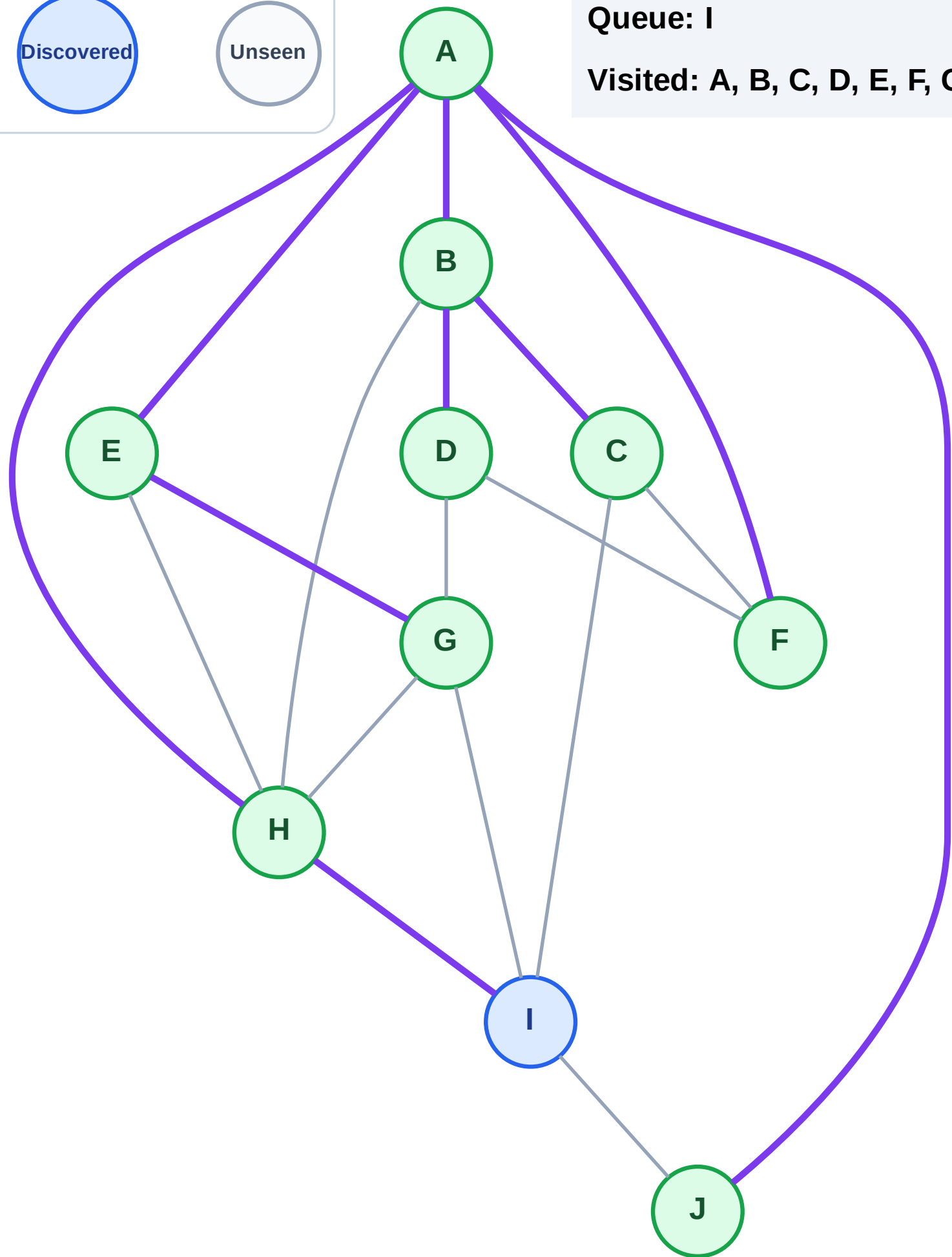
Step 19: No new neighbors from G

Legend



Queue: I

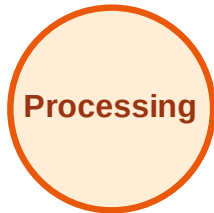
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

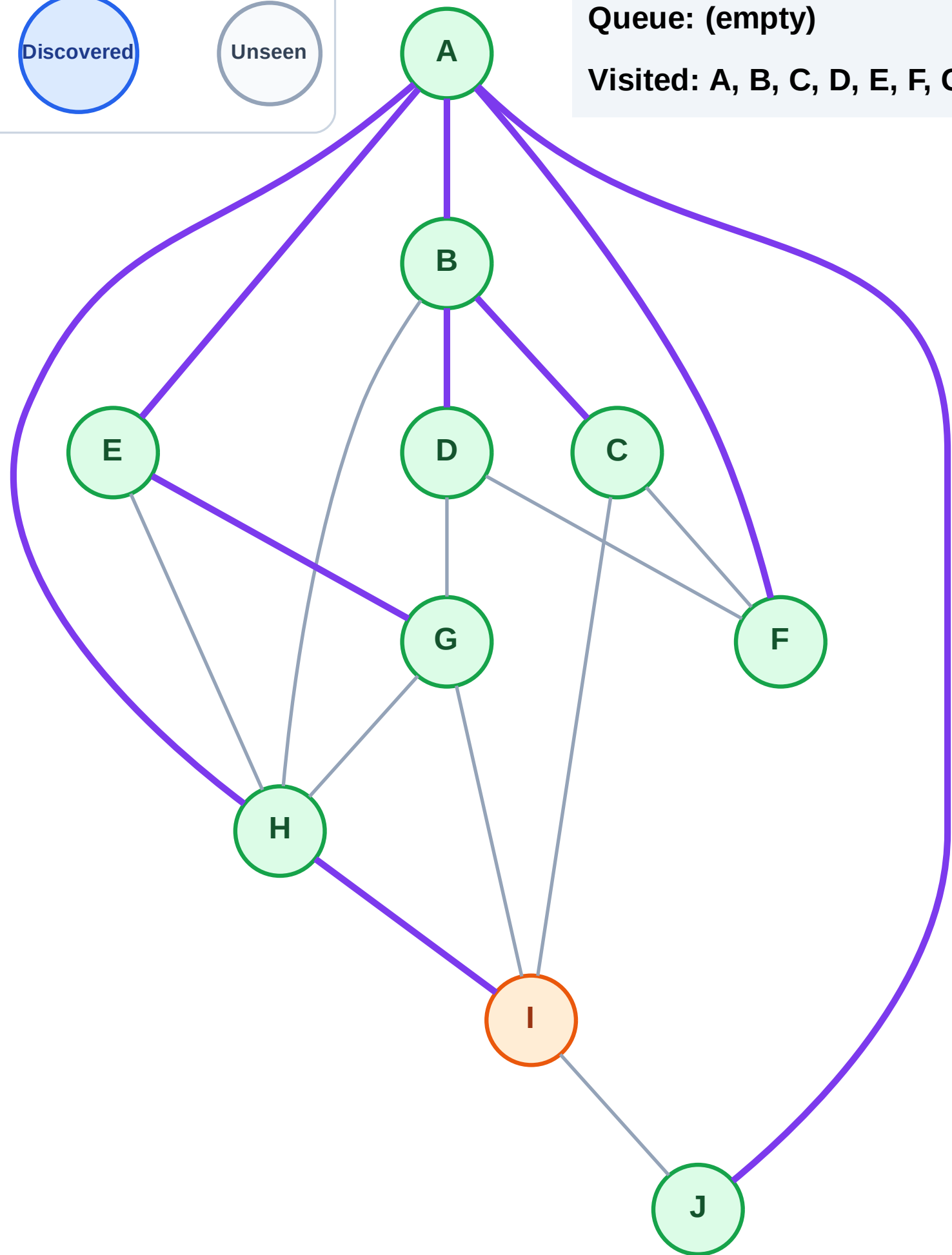
Step 20: Process node I

Legend



Queue: (empty)

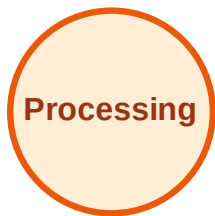
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

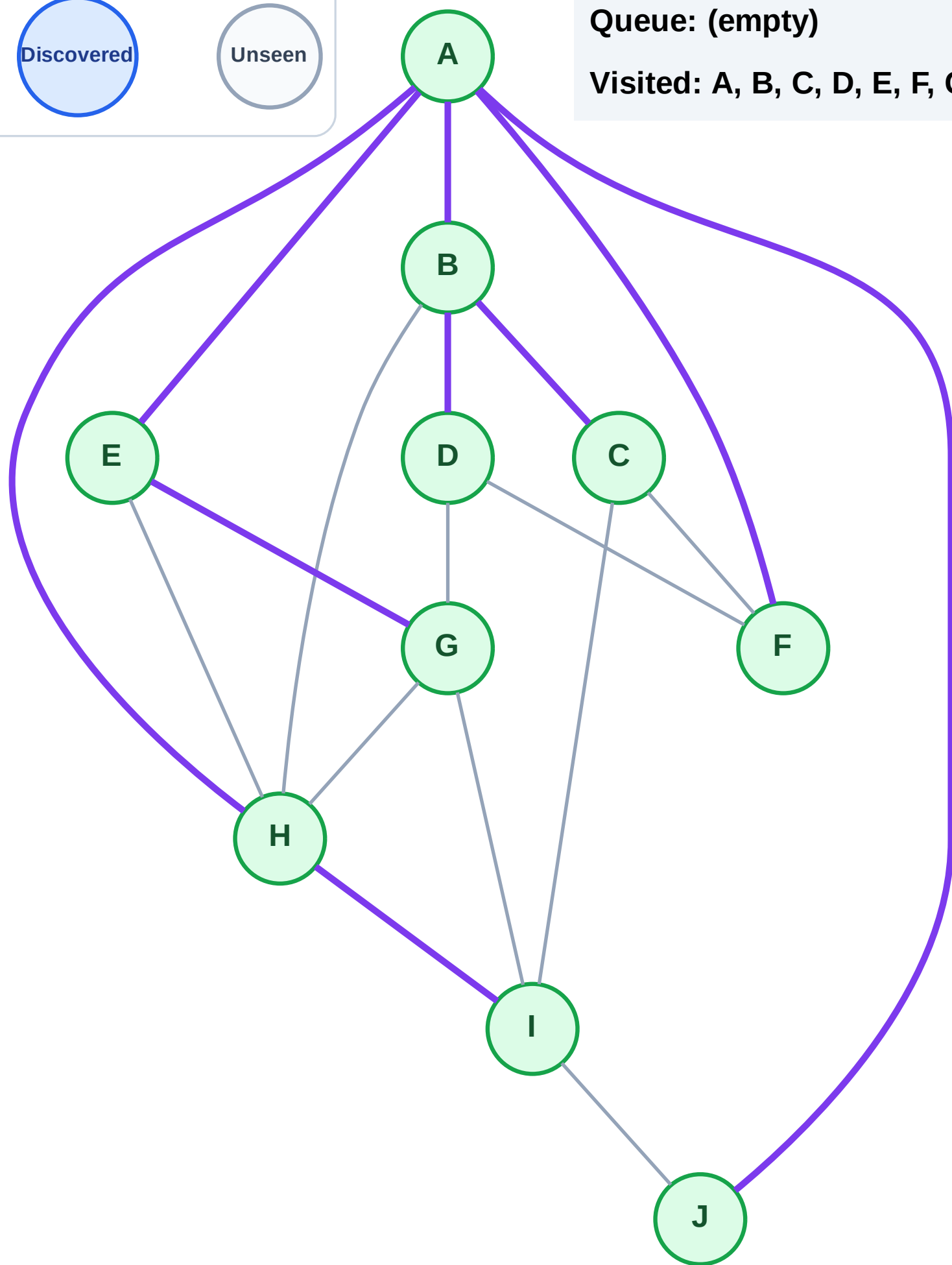
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

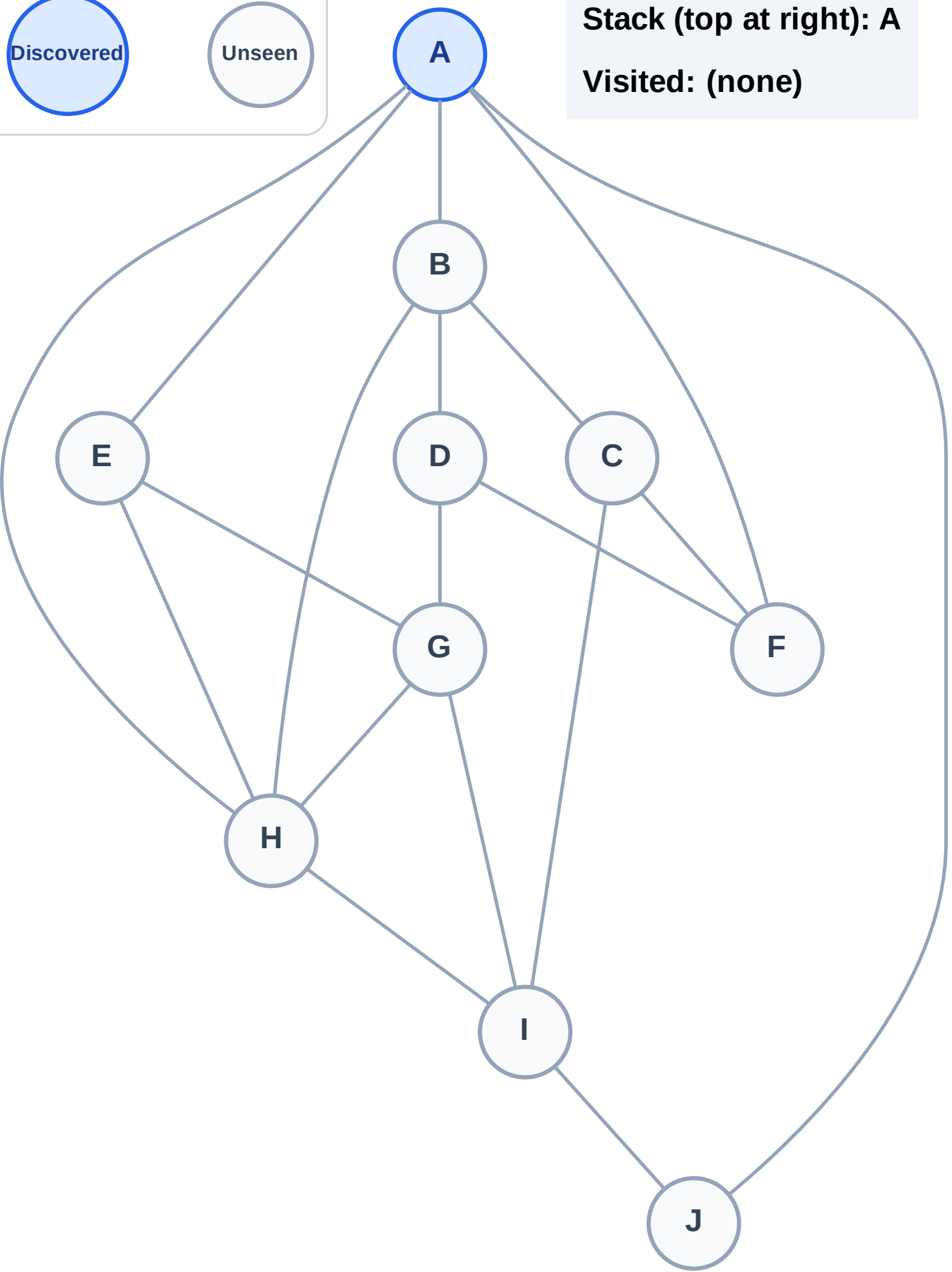
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

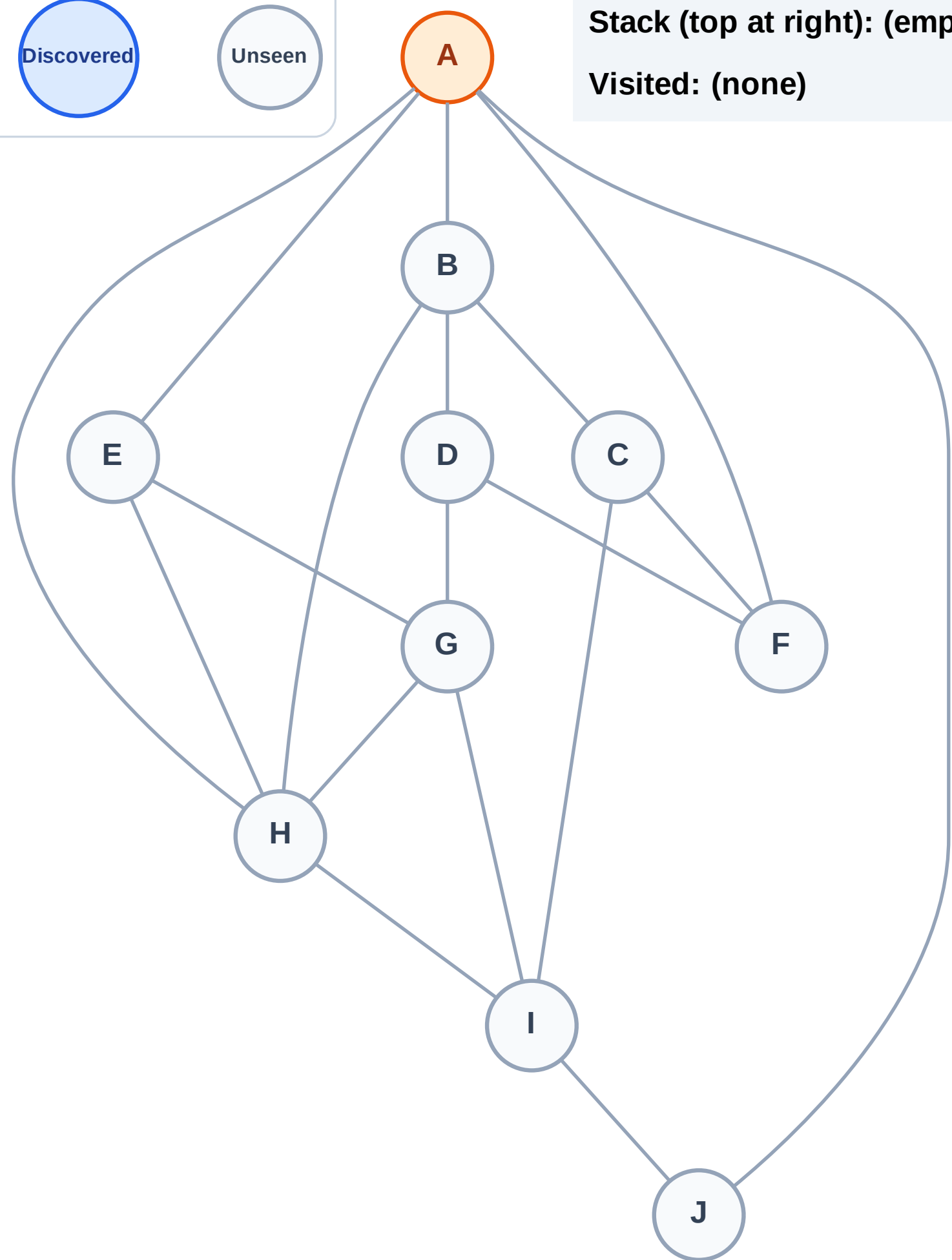
Processing

Discovered

Unseen

Stack (top at right): (empty)

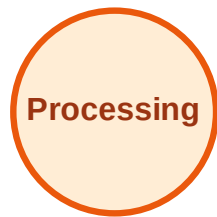
Visited: (none)



DFS Traversal

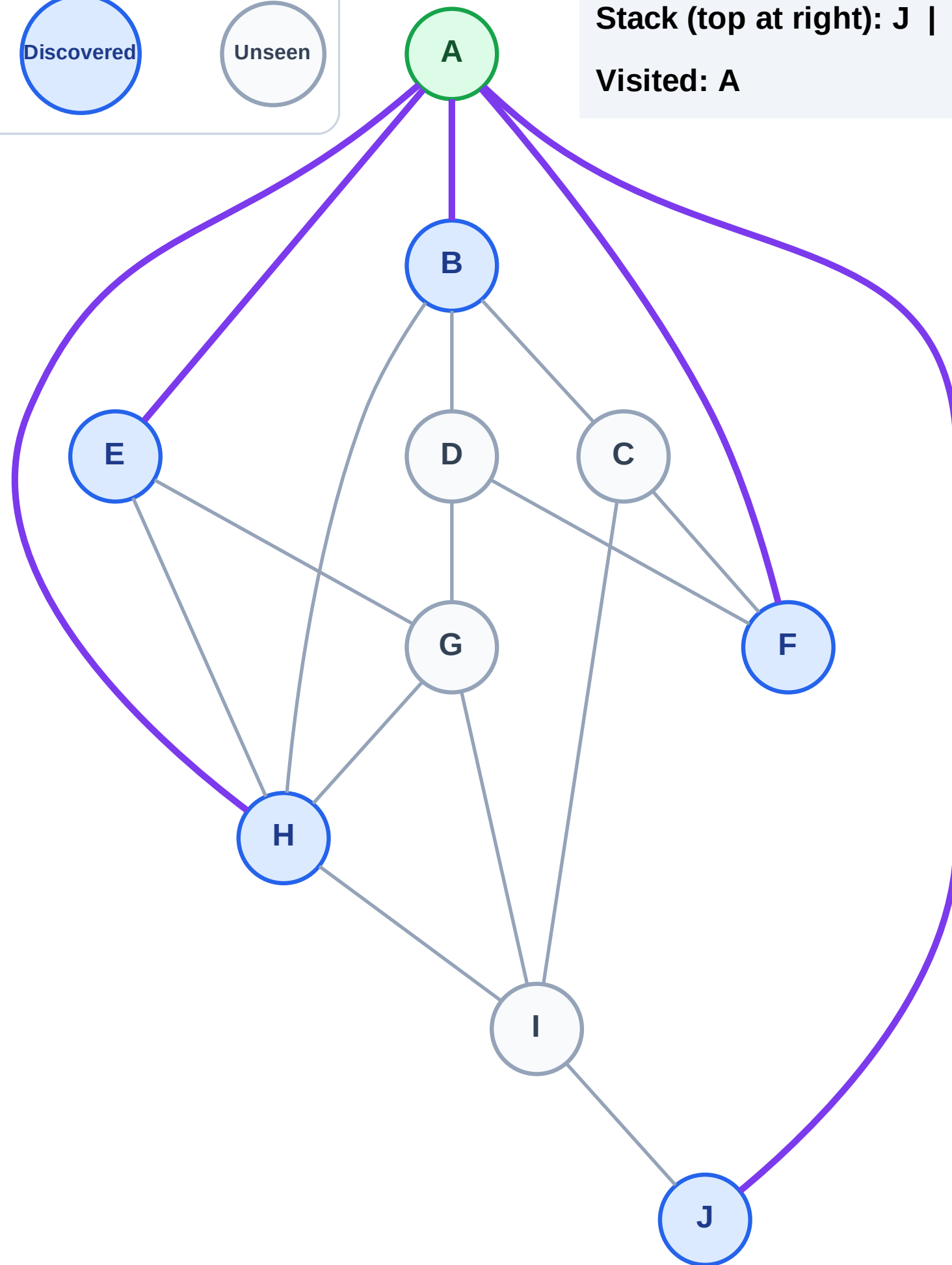
Step 3: Discover J, H, F, E, B from A

Legend



Stack (top at right): J | H | F | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

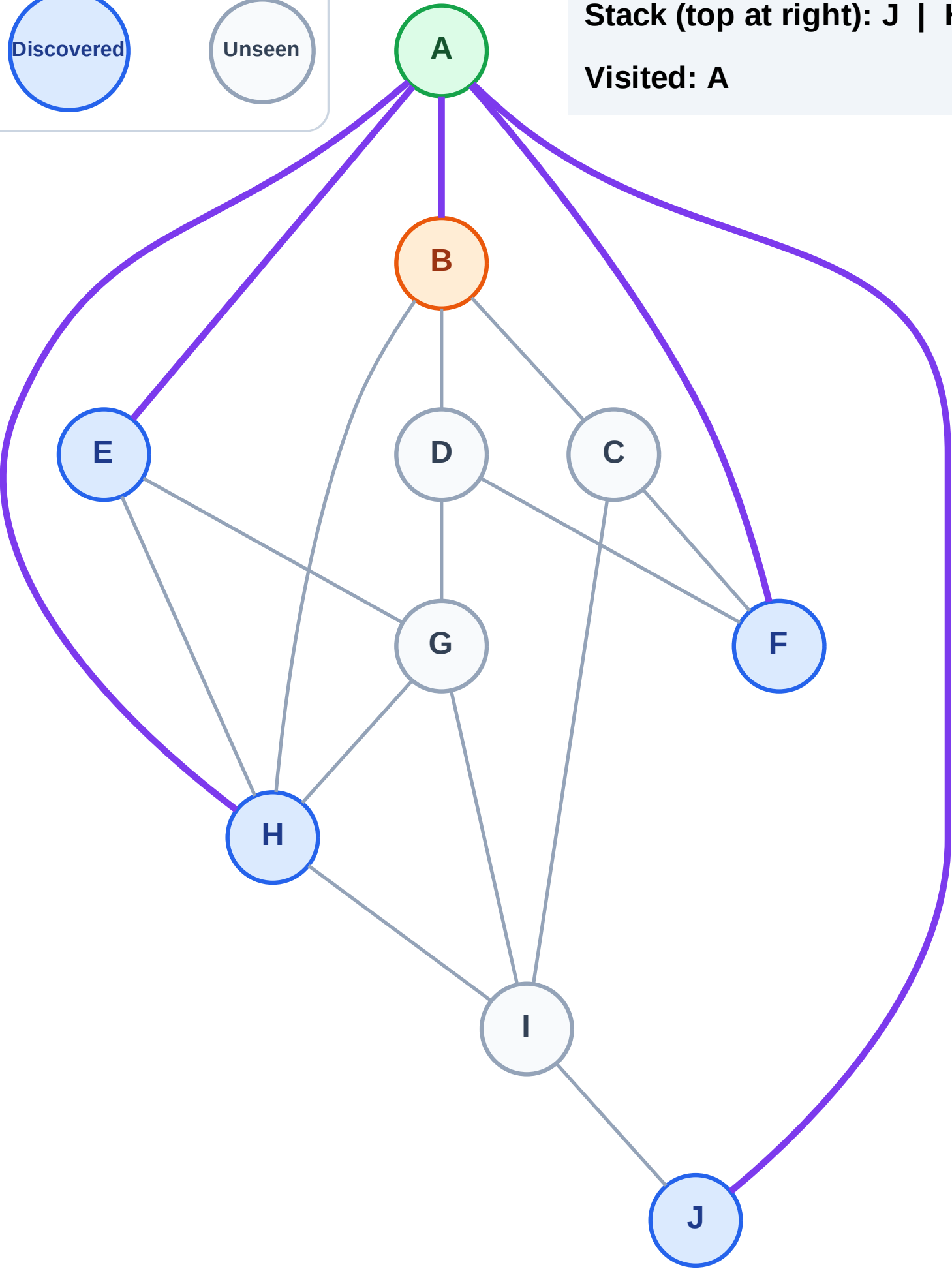
Complete

Processing

Discovered

Unseen

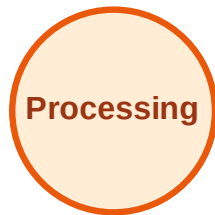
Stack (top at right): J | H | F | E
Visited: A



DFS Traversal

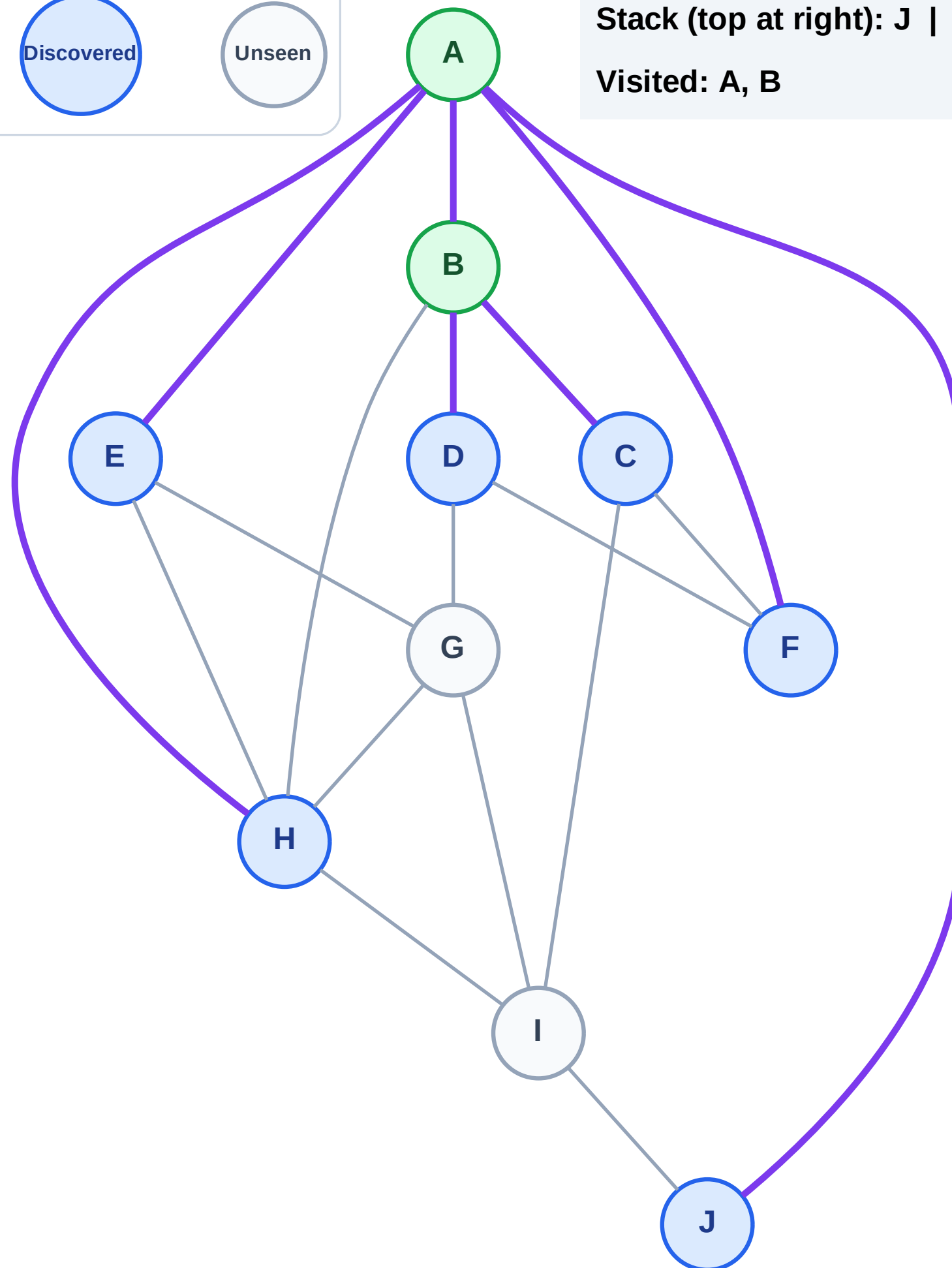
Step 5: Discover D, C from B

Legend



Stack (top at right): J | H | F | E | D | C

Visited: A, B



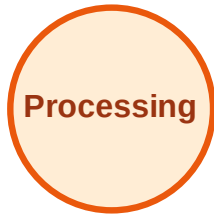
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



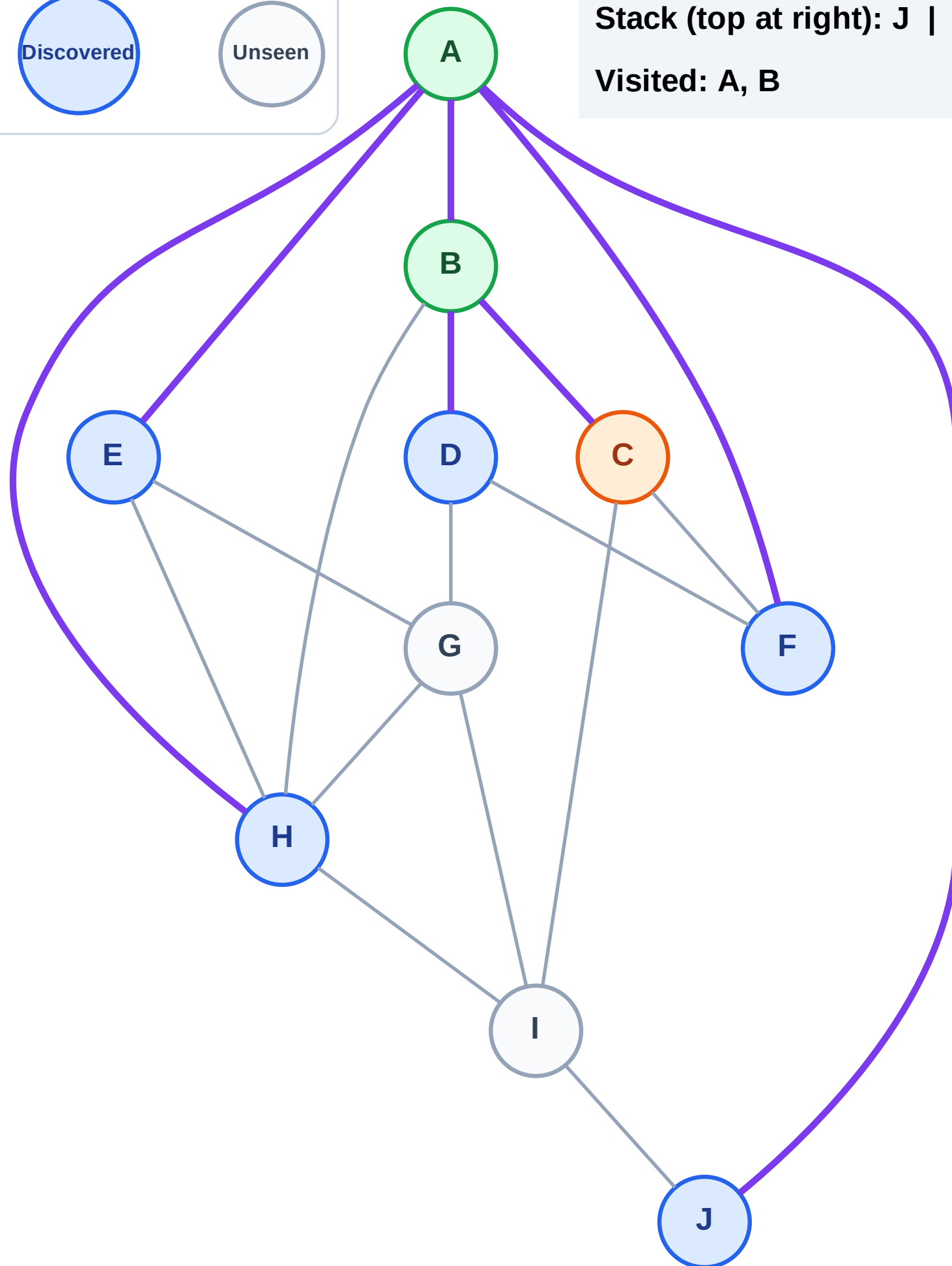
Discovered



Unseen

Stack (top at right): J | H | F | E | D

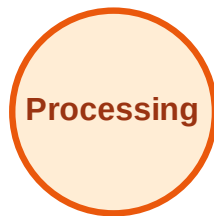
Visited: A, B



DFS Traversal

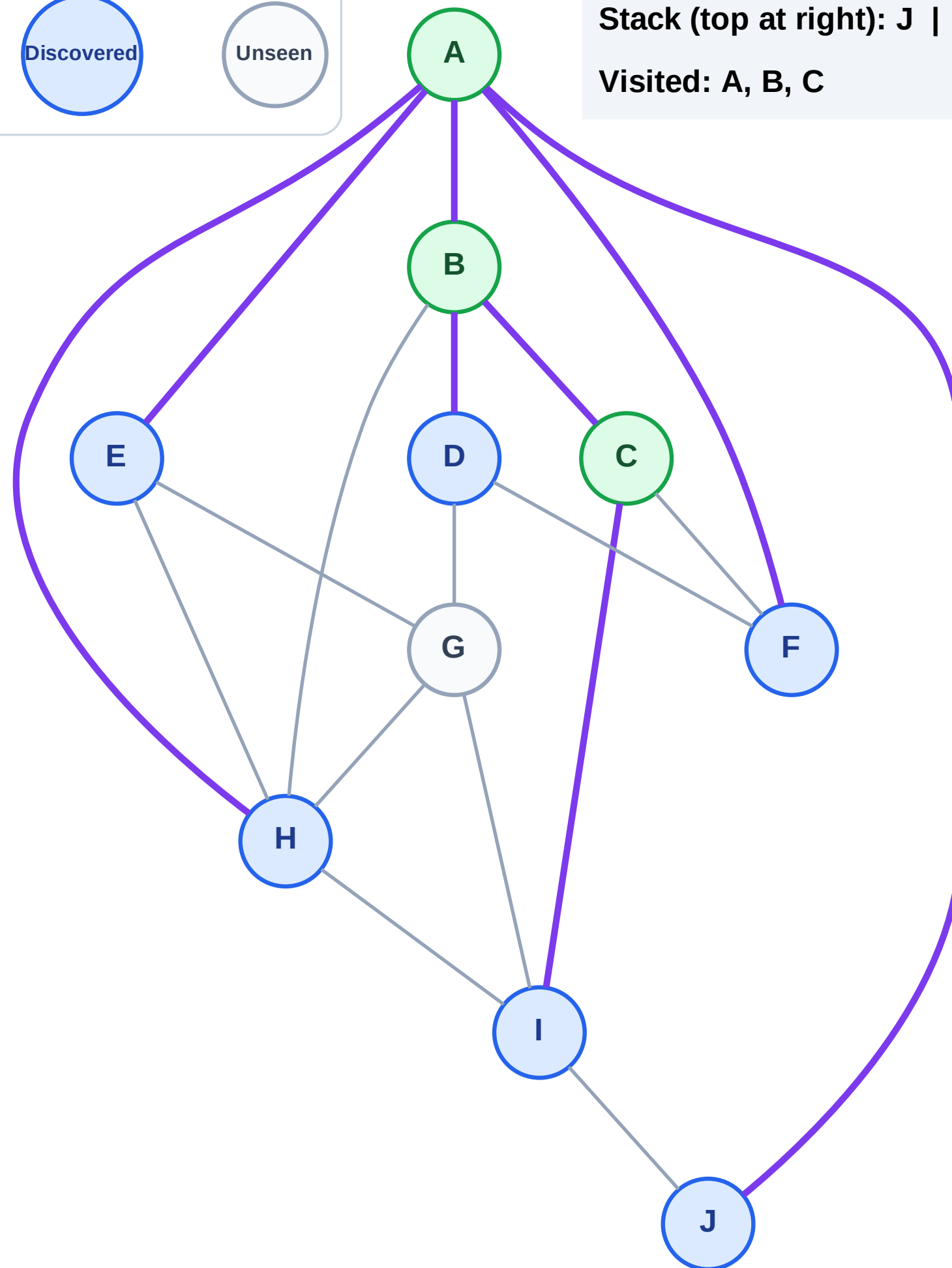
Step 7: Discover I from C

Legend



Stack (top at right): J | H | F | E | D | I

Visited: A, B, C



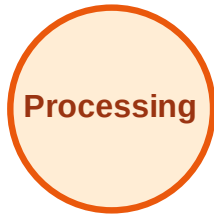
DFS Traversal

Step 8: Process node I

Legend



Complete



Processing



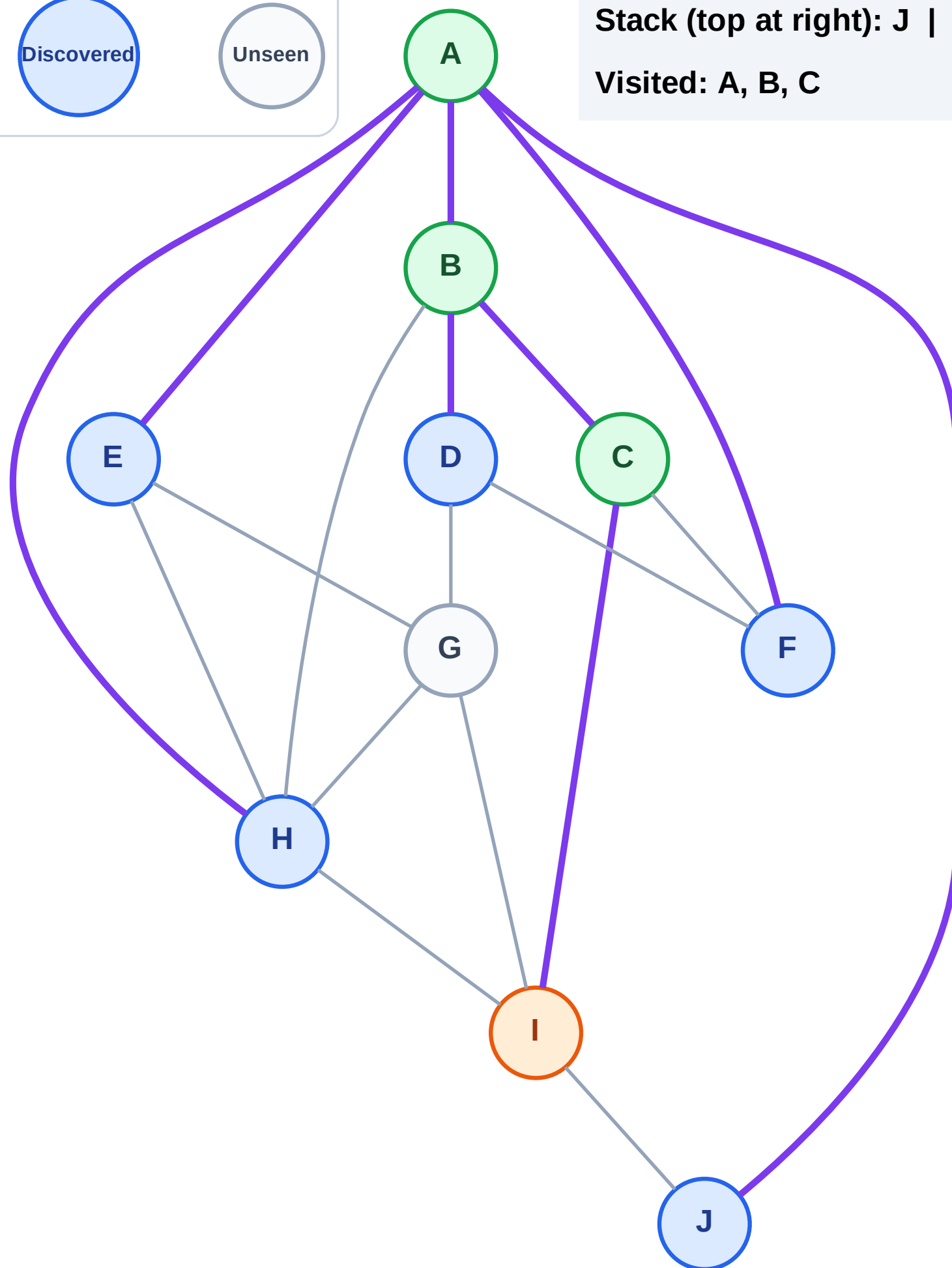
Discovered



Unseen

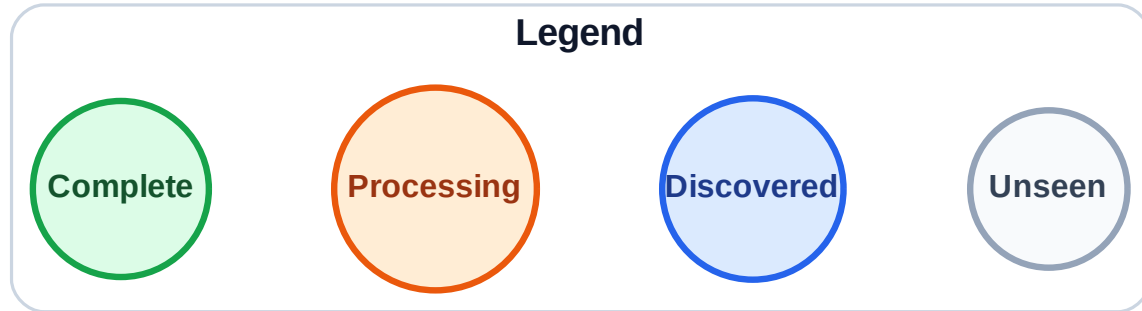
Stack (top at right): J | H | F | E | D

Visited: A, B, C



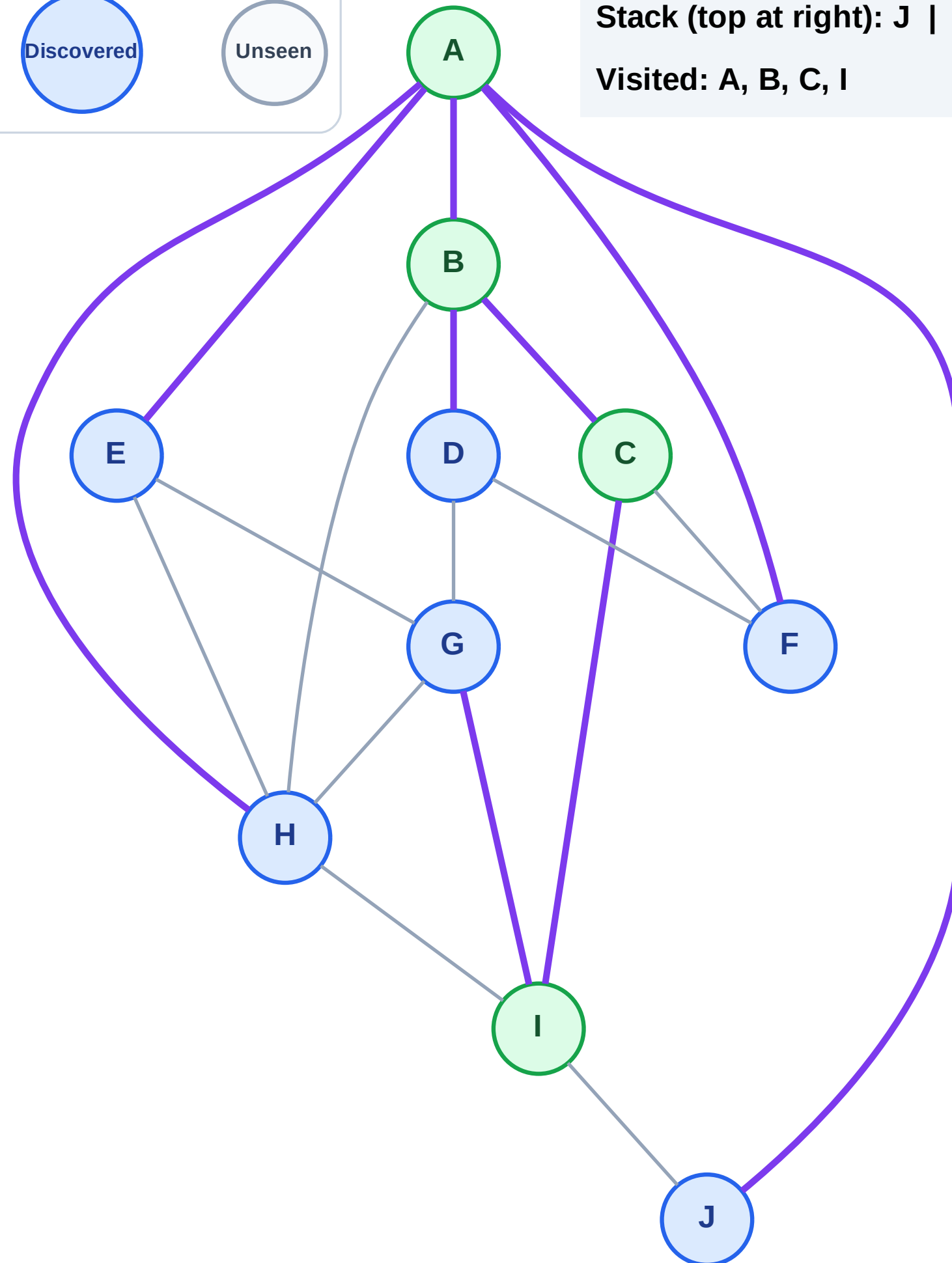
Step 9: Discover G from I

Step 9: Discover G from I



Stack (top at right): J | H | F | E | D | G

Visited: A, B, C, I



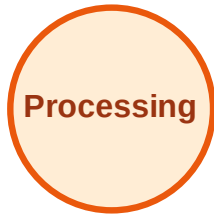
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



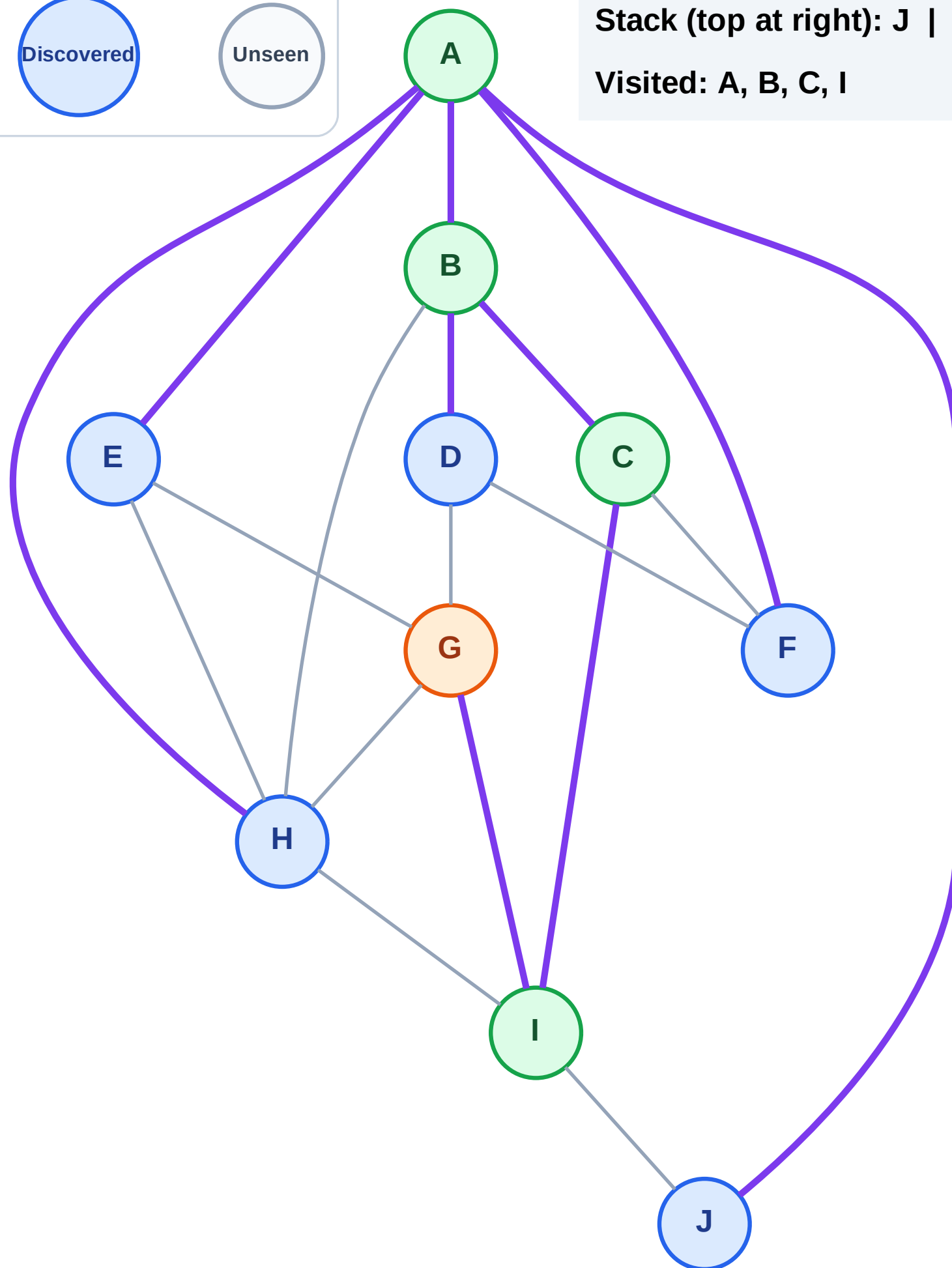
Discovered



Unseen

Stack (top at right): J | H | F | E | D

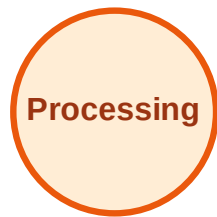
Visited: A, B, C, I



DFS Traversal

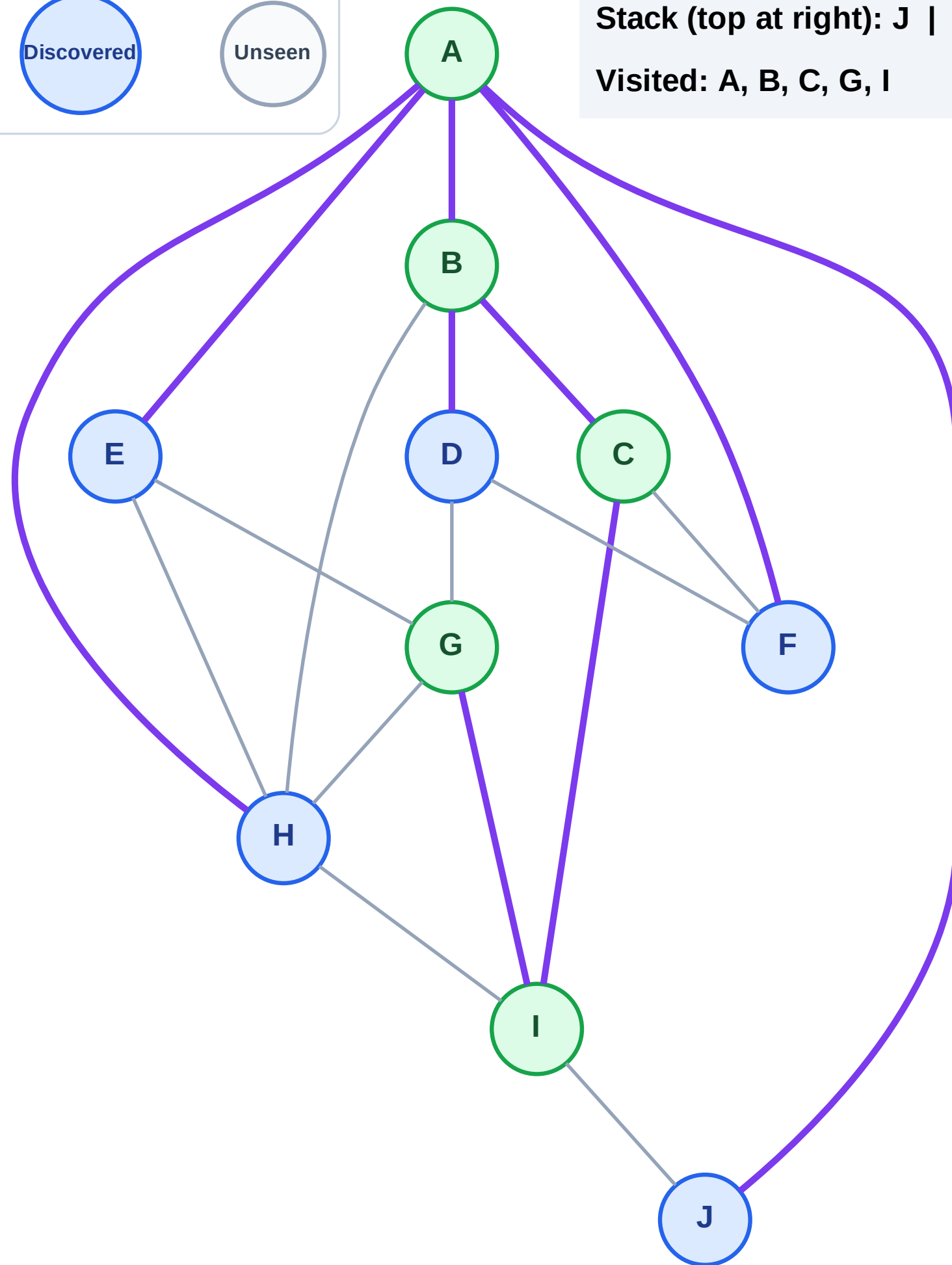
Step 11: No new neighbors from G

Legend



Stack (top at right): J | H | F | E | D

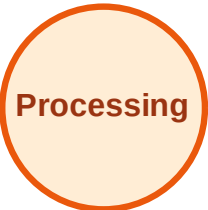
Visited: A, B, C, G, I



DFS Traversal

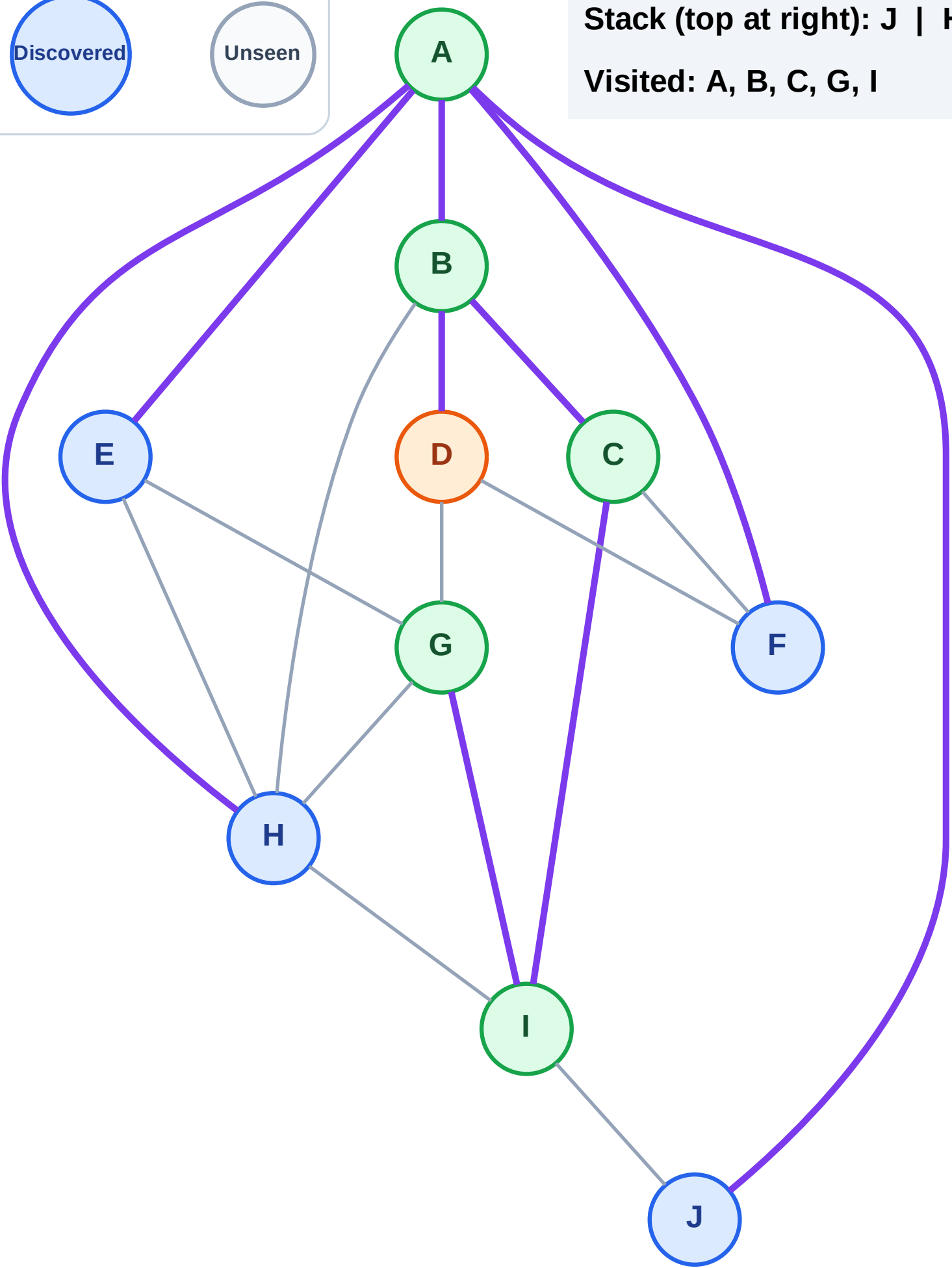
Step 12: Process node D

Legend



Stack (top at right): J | H | F | E

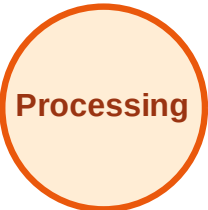
Visited: A, B, C, G, I



DFS Traversal

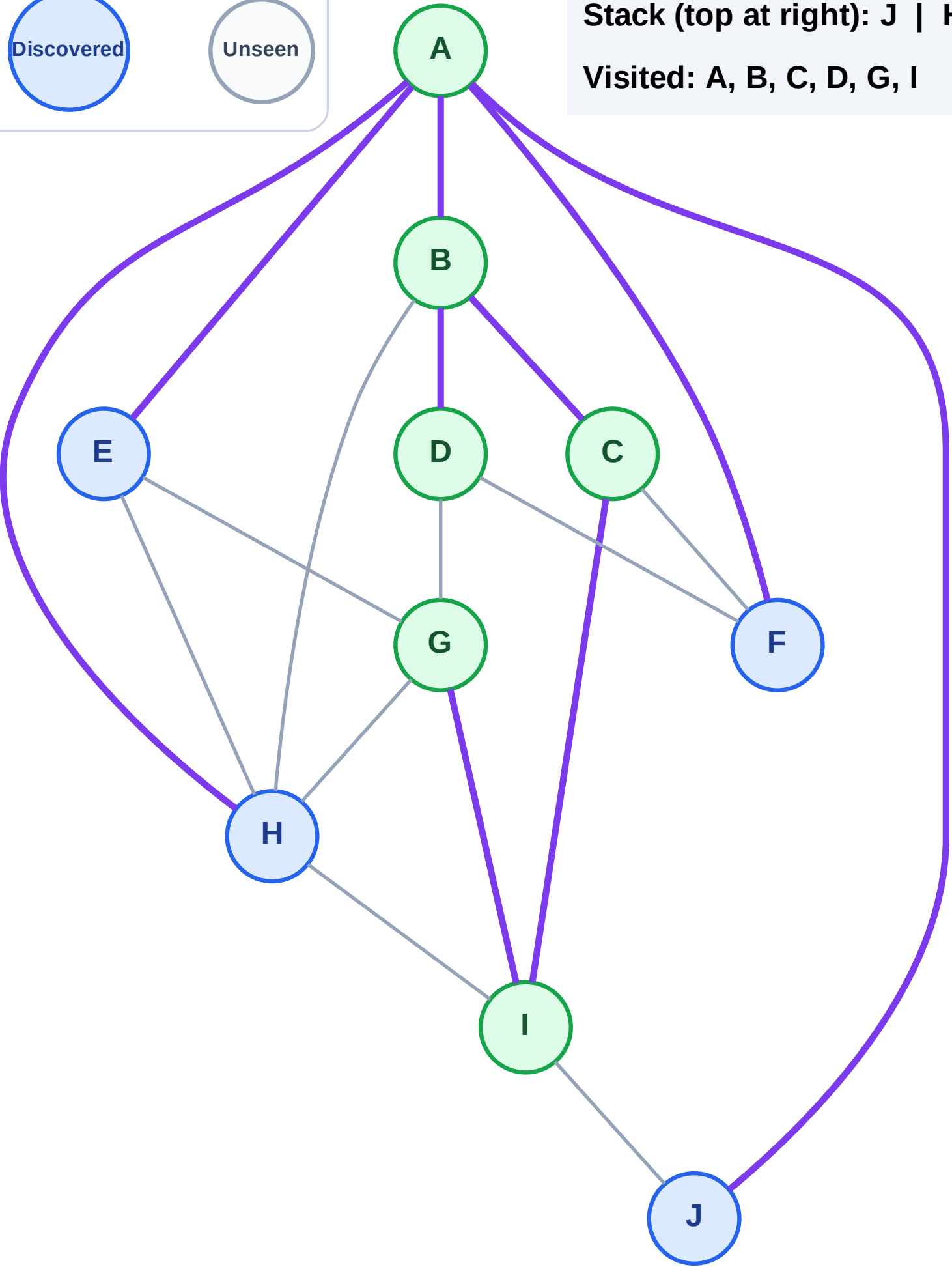
Step 13: No new neighbors from D

Legend



Stack (top at right): J | H | F | E

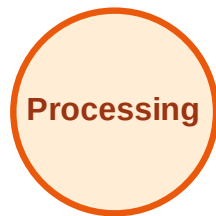
Visited: A, B, C, D, G, I



DFS Traversal

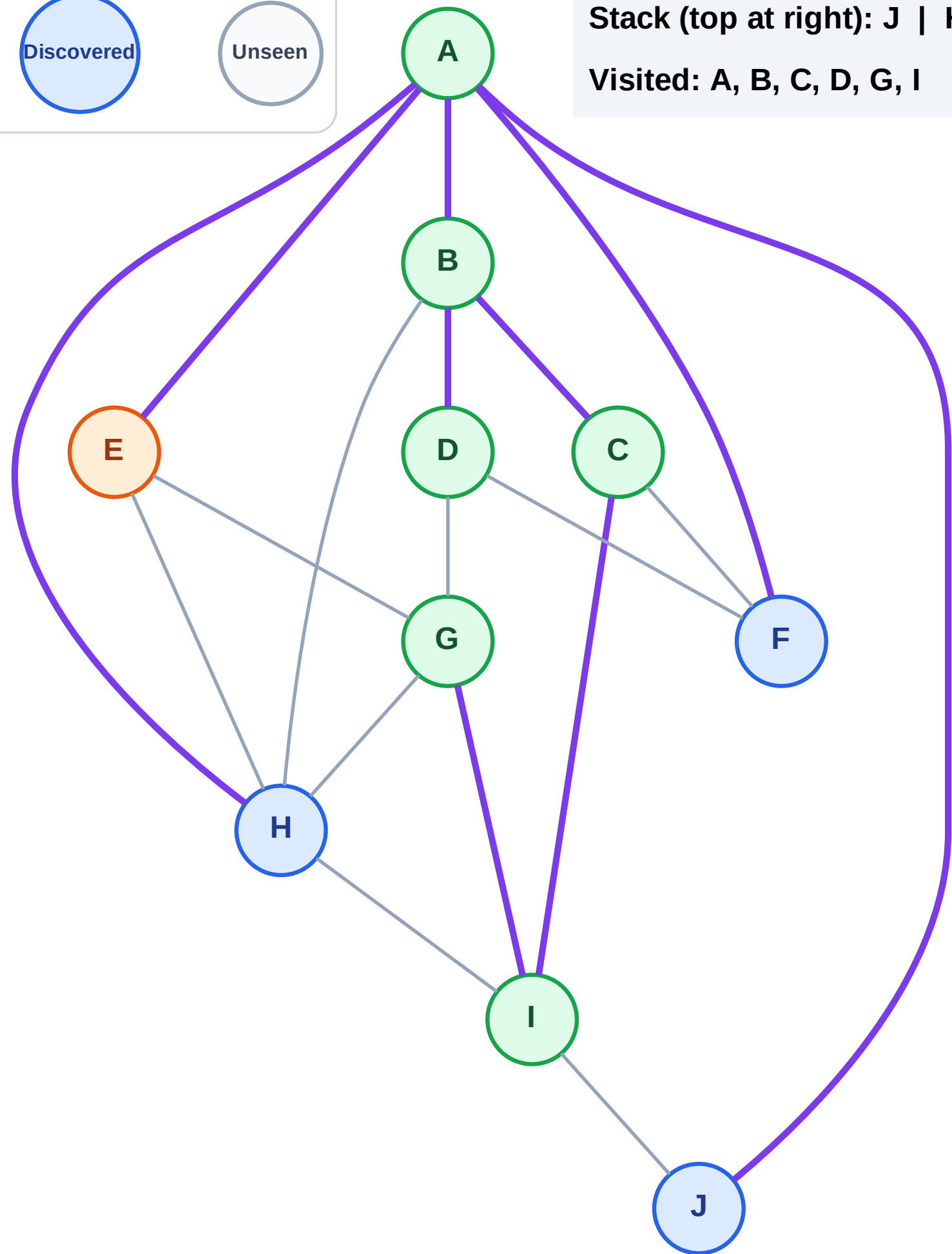
Step 14: Process node E

Legend



Stack (top at right): J | H | F

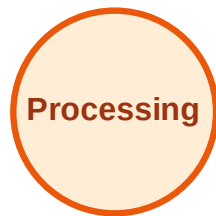
Visited: A, B, C, D, G, I



DFS Traversal

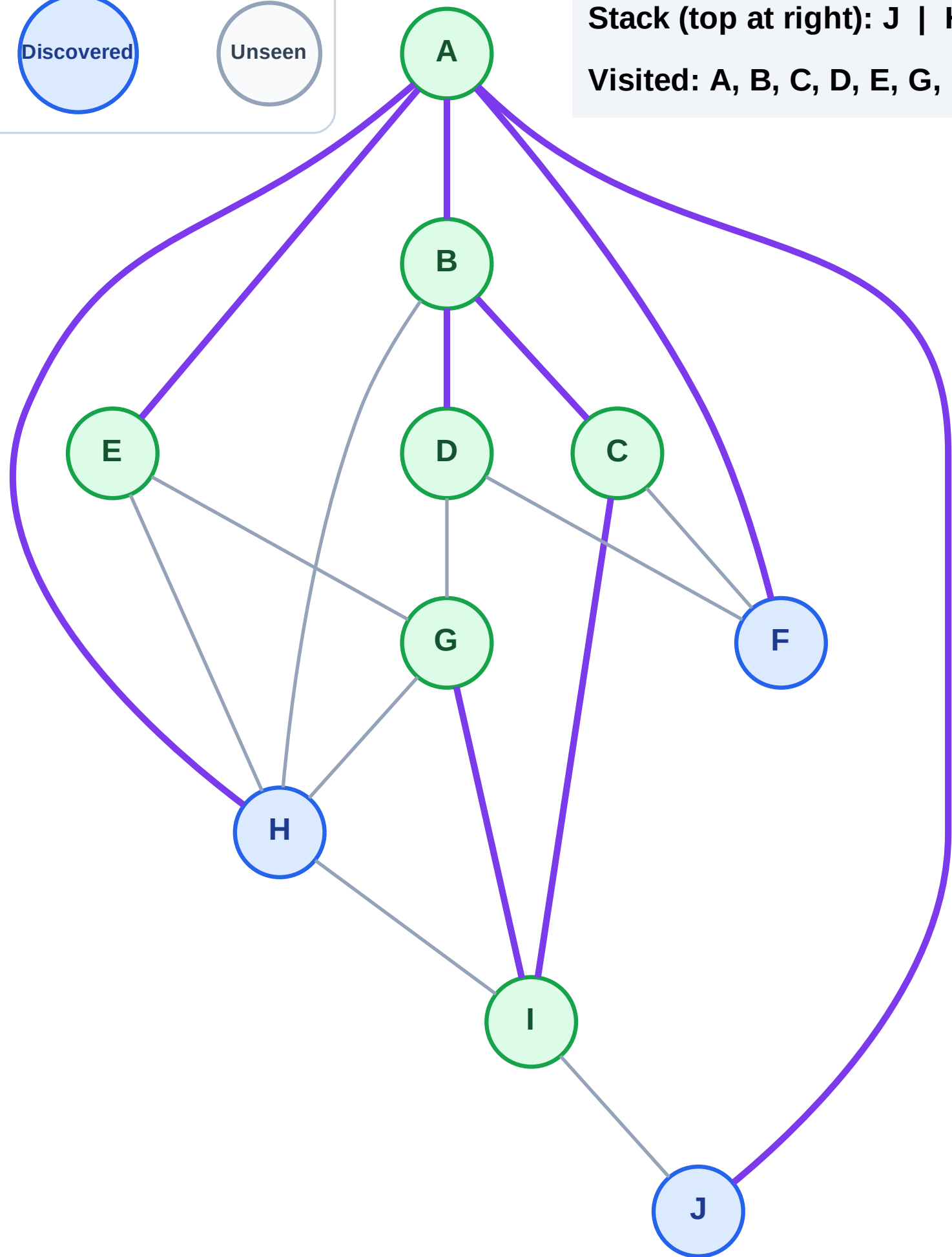
Step 15: No new neighbors from E

Legend



Stack (top at right): J | H | F

Visited: A, B, C, D, E, G, I



DFS Traversal

Step 16: Process node F

Legend

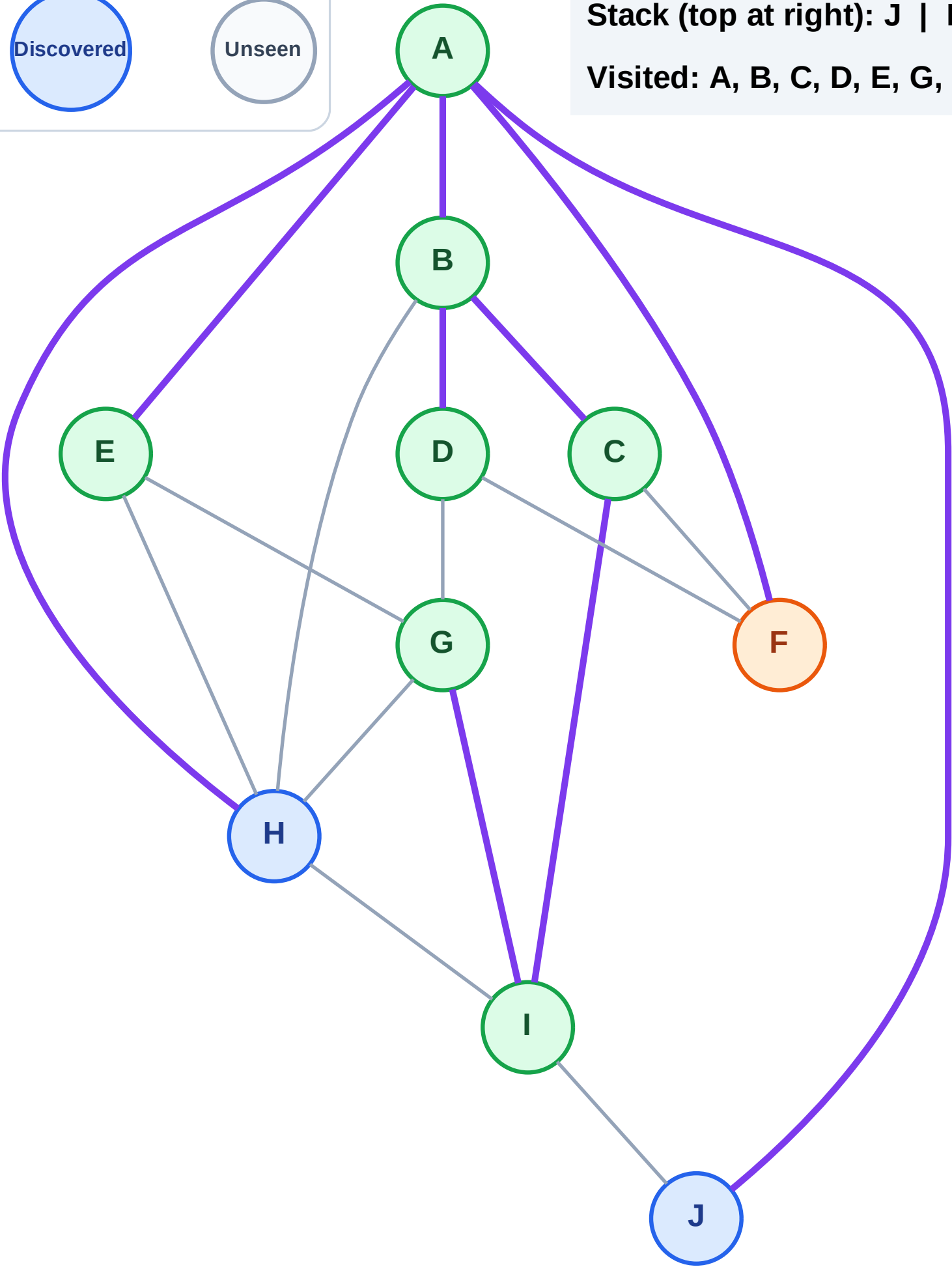
Complete

Processing

Discovered

Unseen

Stack (top at right): J | H
Visited: A, B, C, D, E, G, I



DFS Traversal

Step 17: No new neighbors from F

Legend

Complete

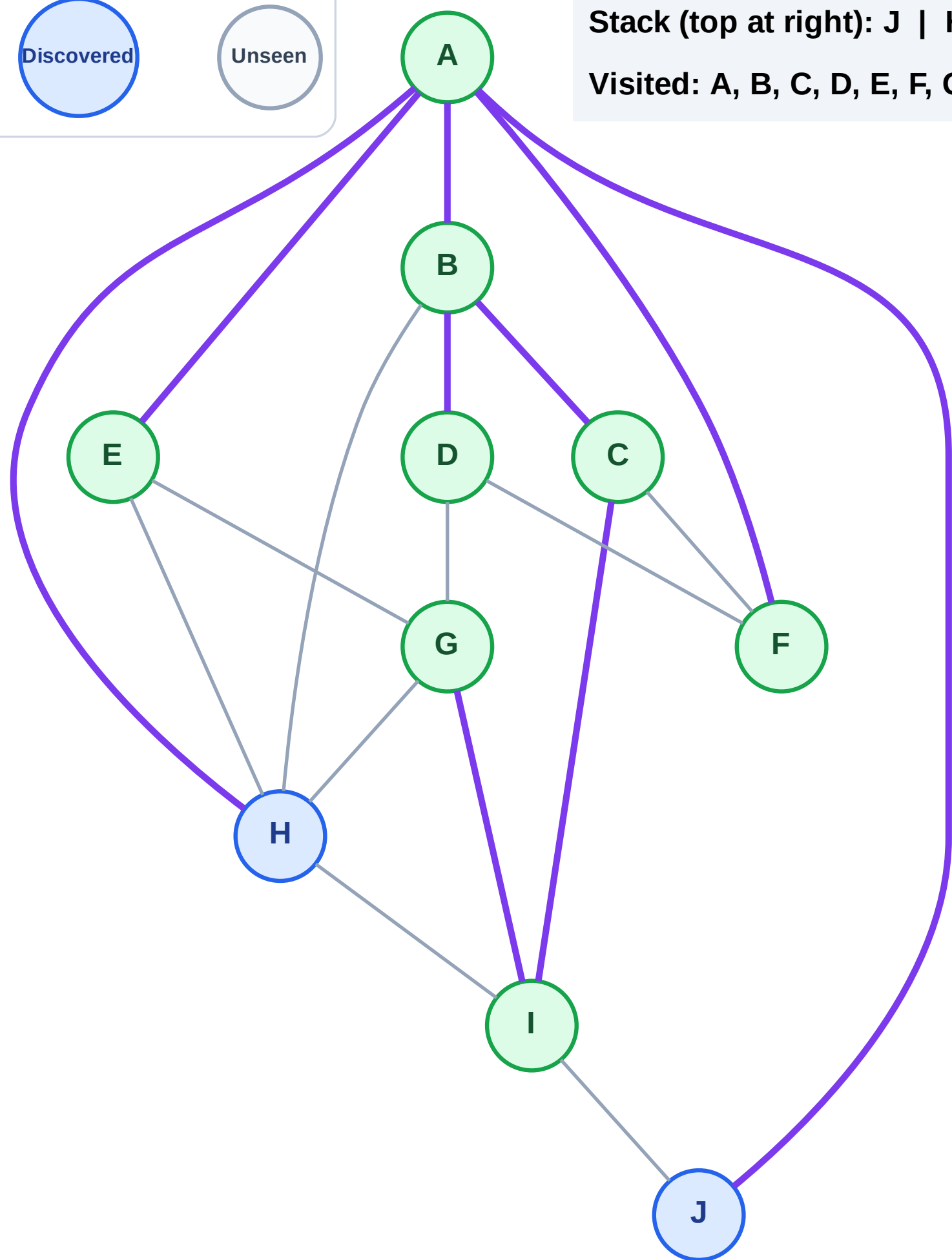
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



DFS Traversal

Step 18: Process node H

Legend

Complete

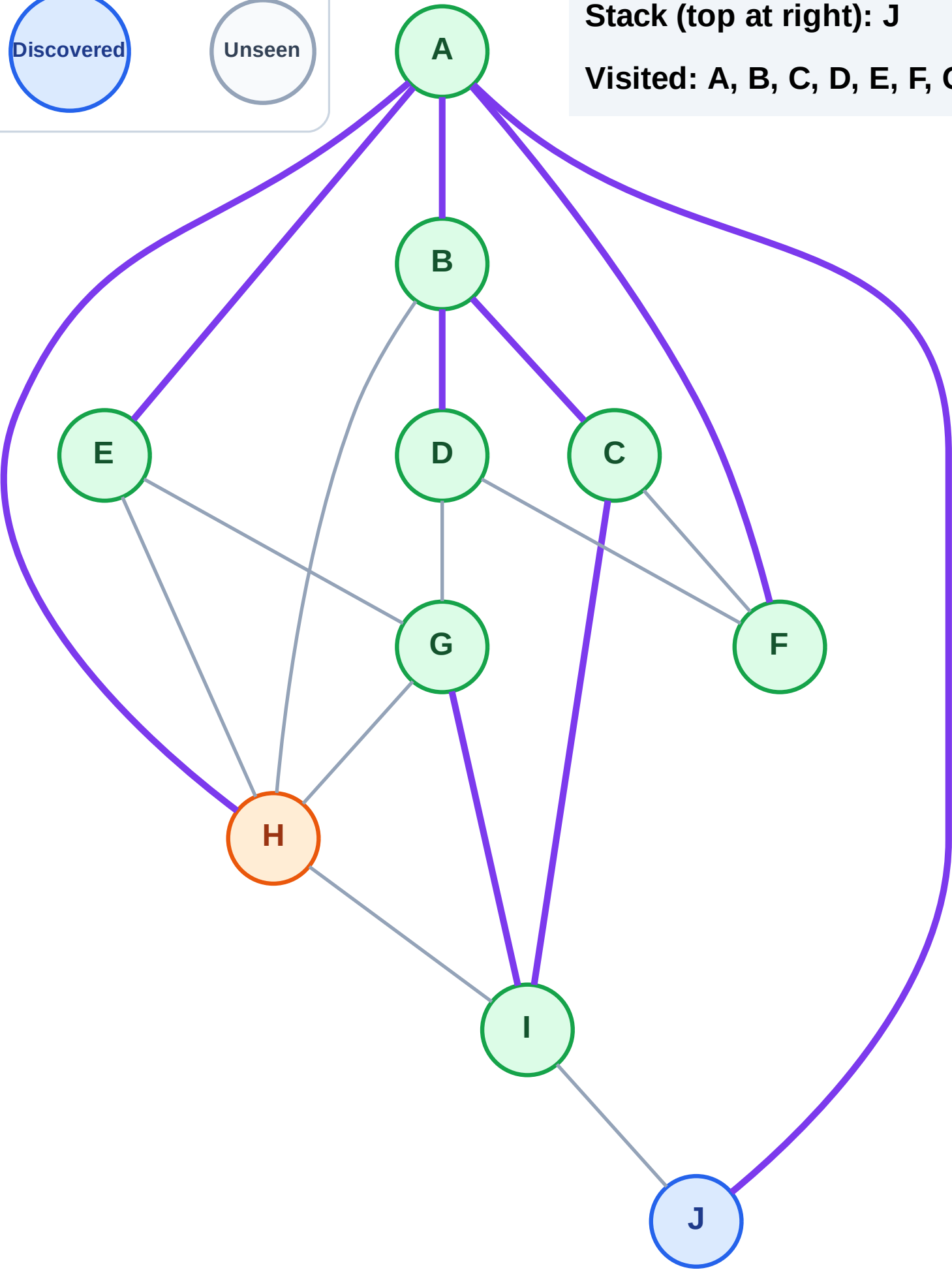
Processing

Discovered

Unseen

Stack (top at right): J

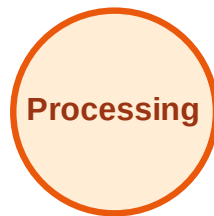
Visited: A, B, C, D, E, F, G, I



DFS Traversal

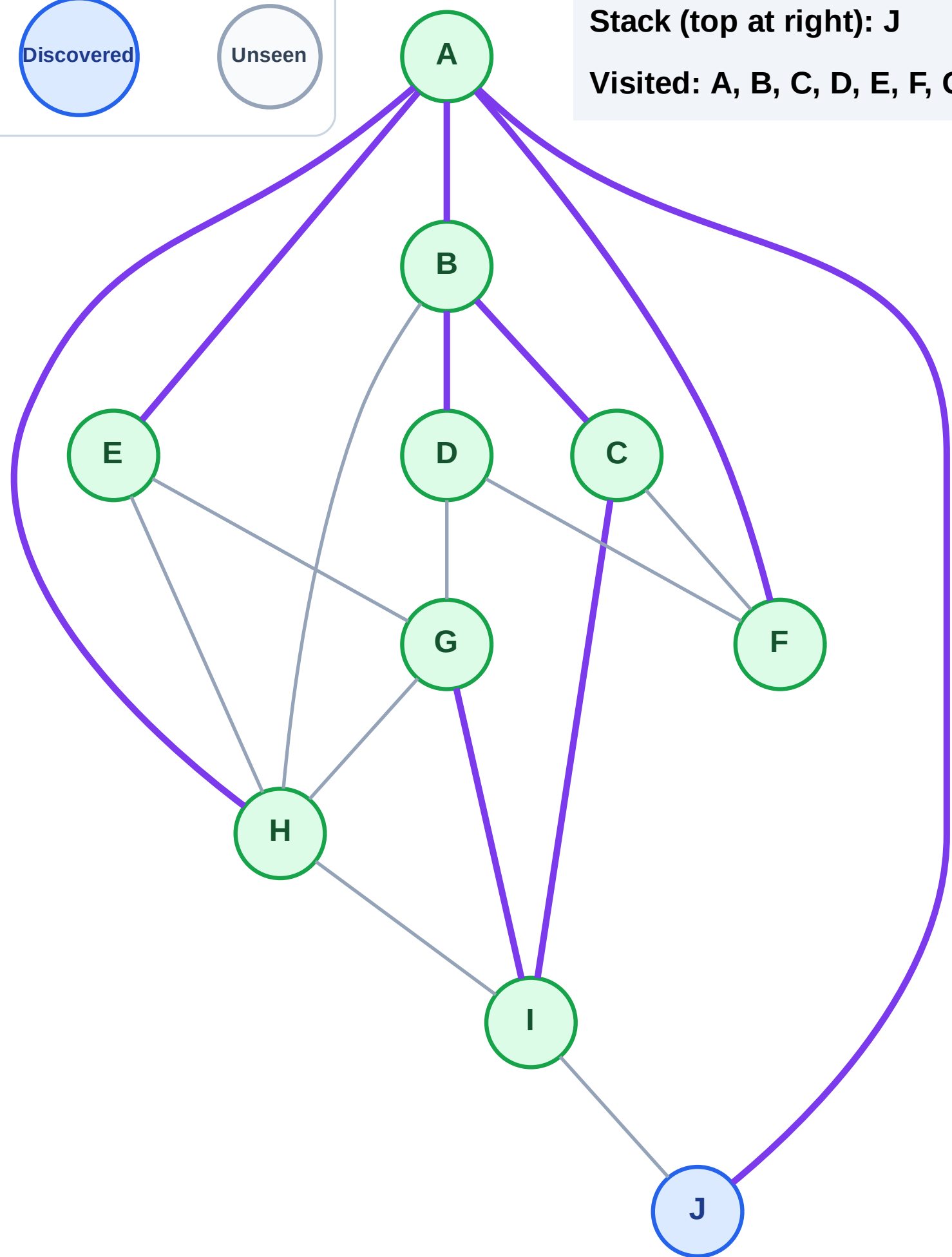
Step 19: No new neighbors from H

Legend



Stack (top at right): J

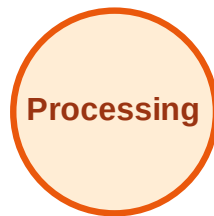
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

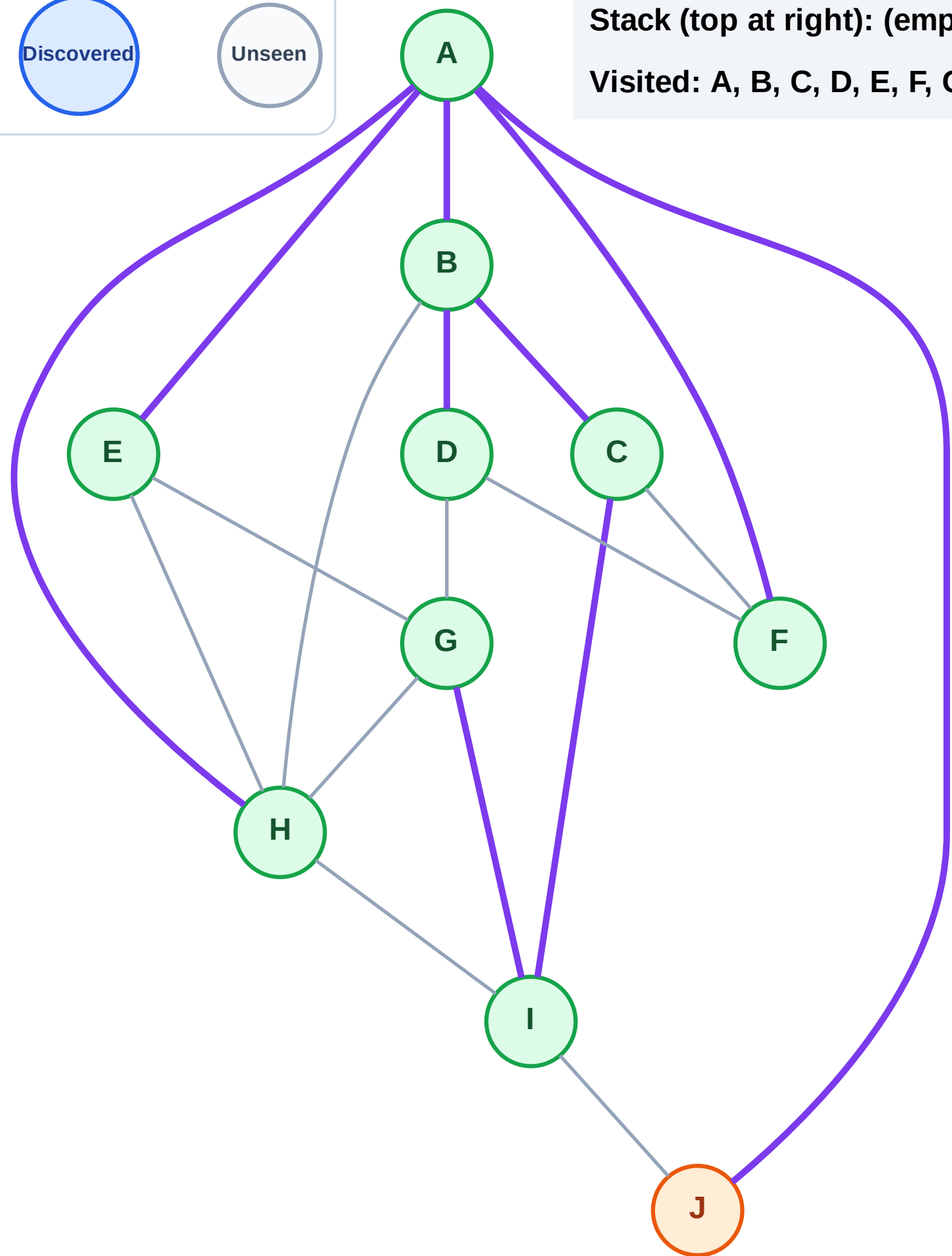
Step 20: Process node J

Legend



Stack (top at right): (empty)

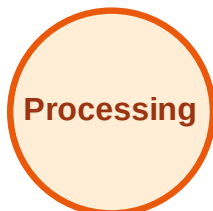
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

